Software Engineering Lab

Crystal Reports – Part 1

Lab 7

Agenda

- Introduction to Crystal Reports
- Crystal Reports Creation
- Calling Crystal Reports From C# Form
- Crystal Reports with Parameters
- Passing Parameter Values From C# Form To The Crystal Report

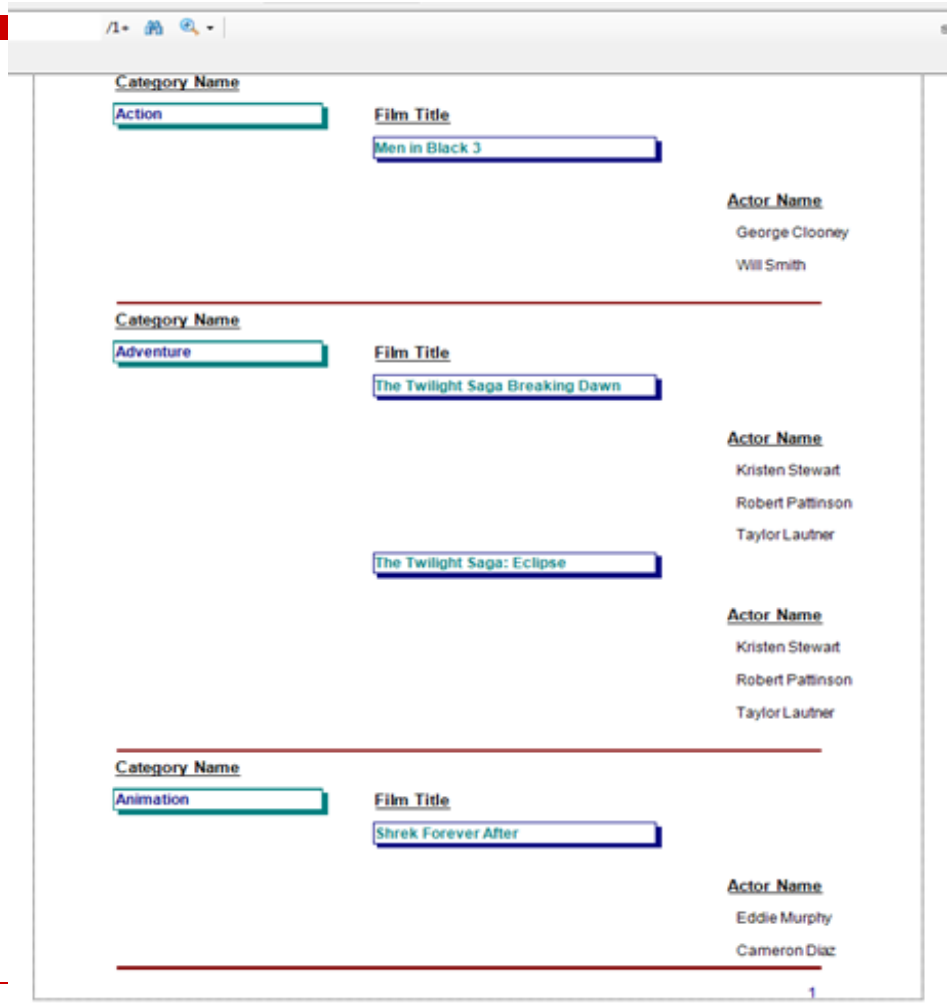
Introduction to Crystal Reports

- Crystal Report is a database reporting application used by many different types of businesses to generate accurate and powerful reports that assist in business decision making.
- It can access and analyze different types of data sources as Oracle, Microsoft SQL Server, etc.

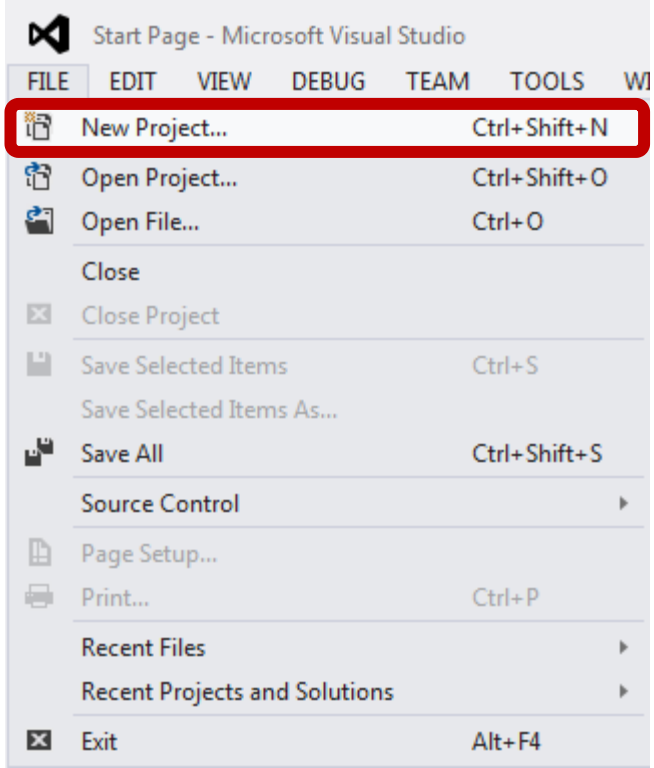
Creating New Crystal Report Application

Crystal Reports Creation

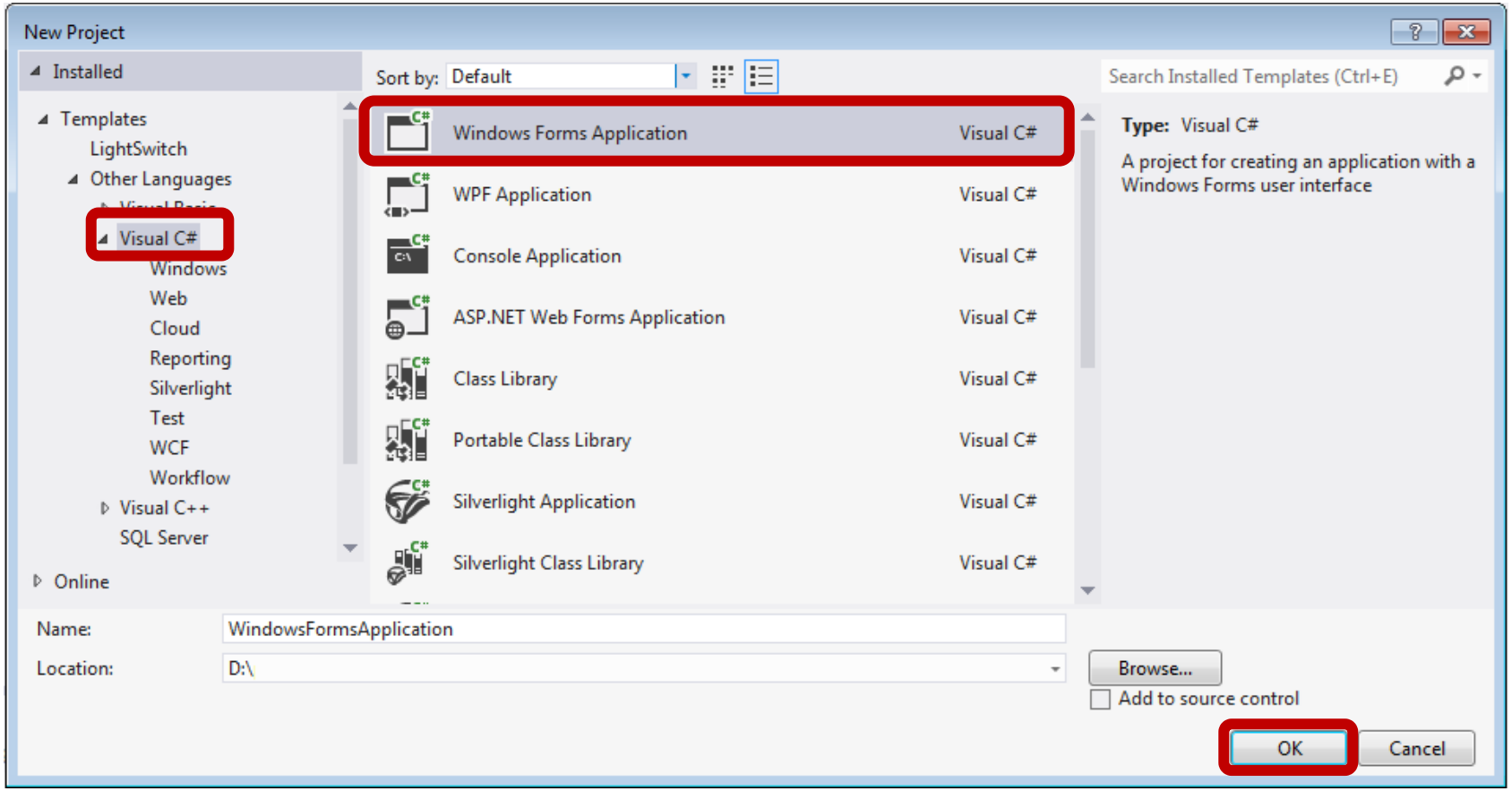
- Create a Report that displays the names of the movies for each category and the names of the actors for each movie title.



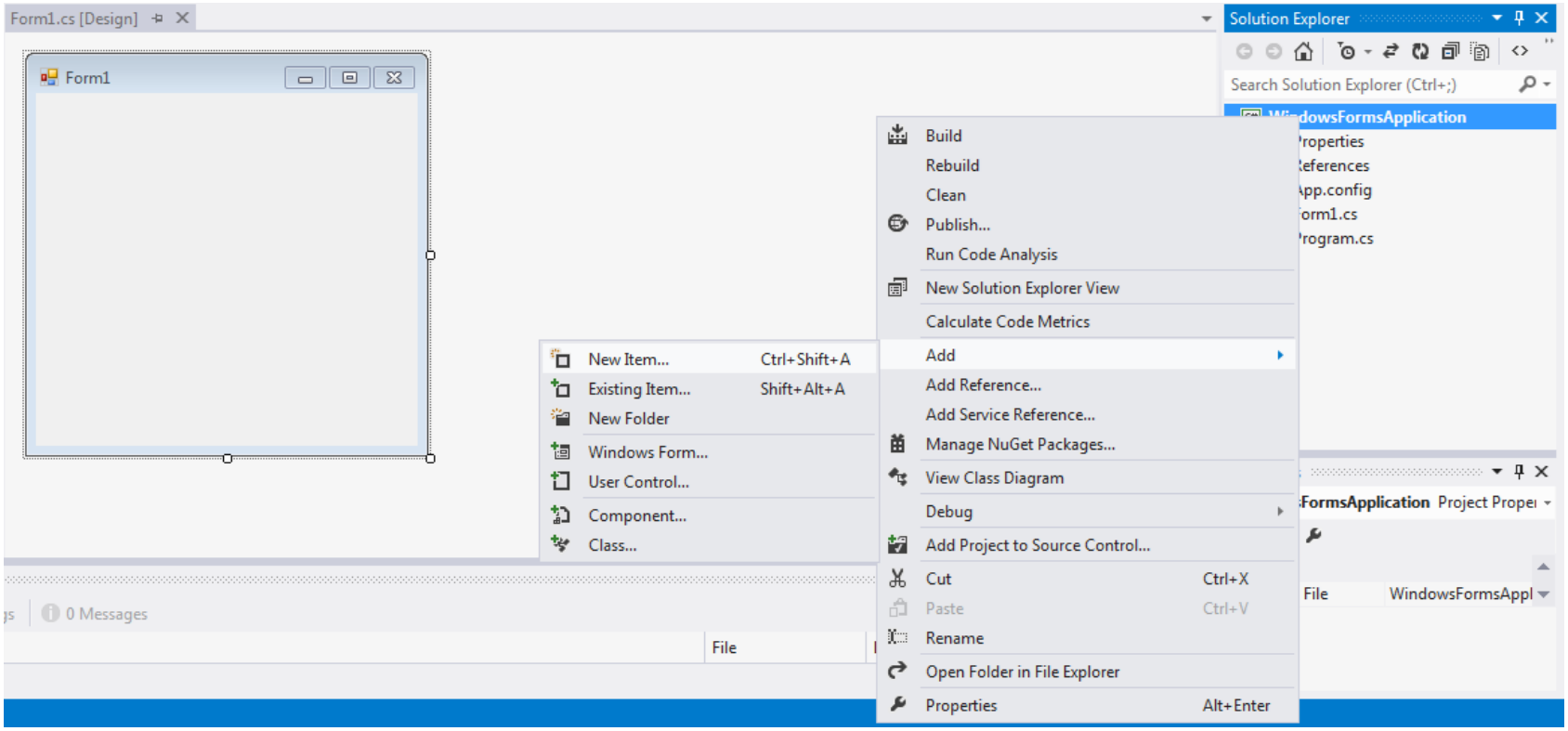
Crystal Reports Creation (cont'd)



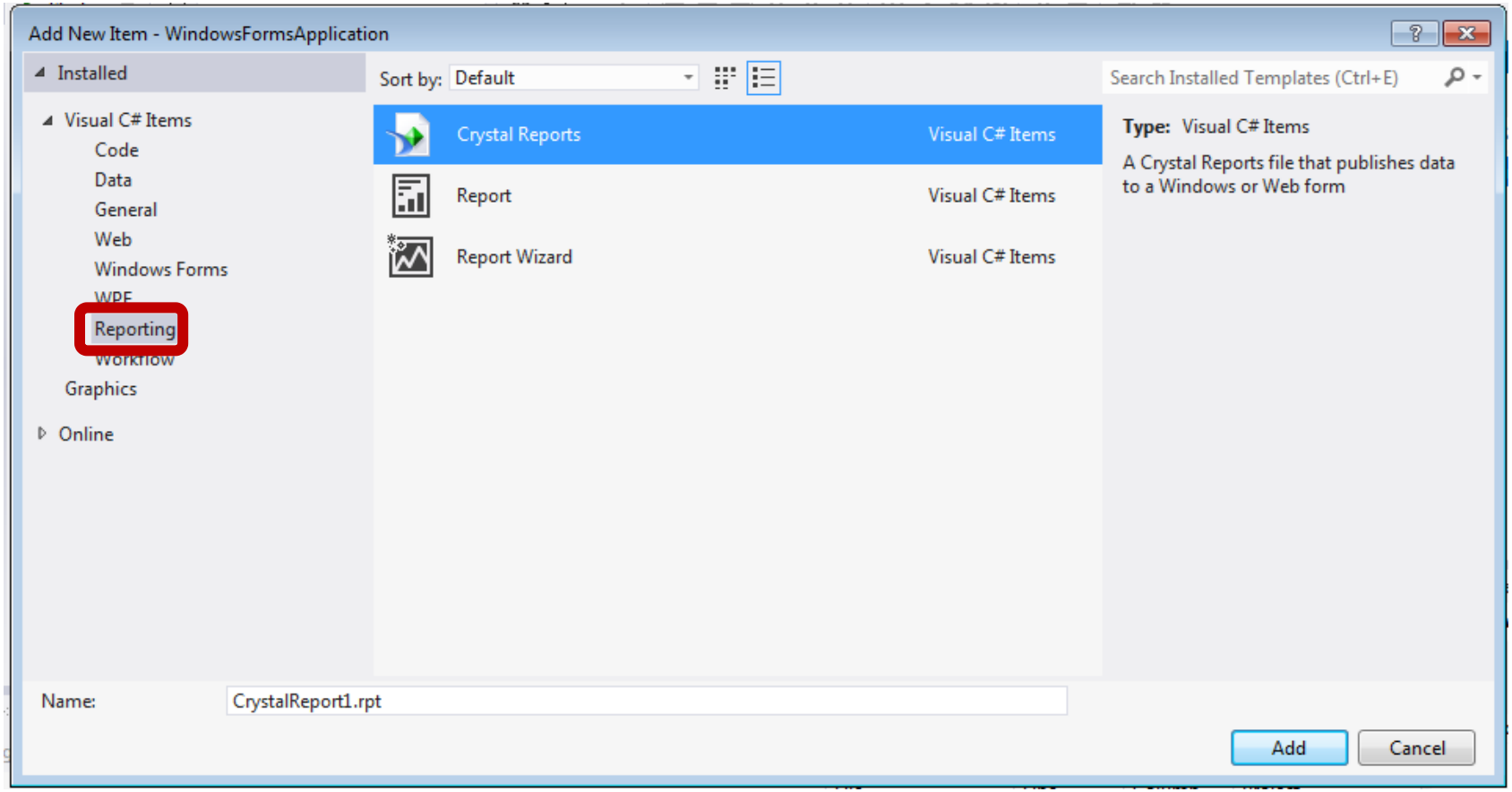
Crystal Reports Creation(cont'd)



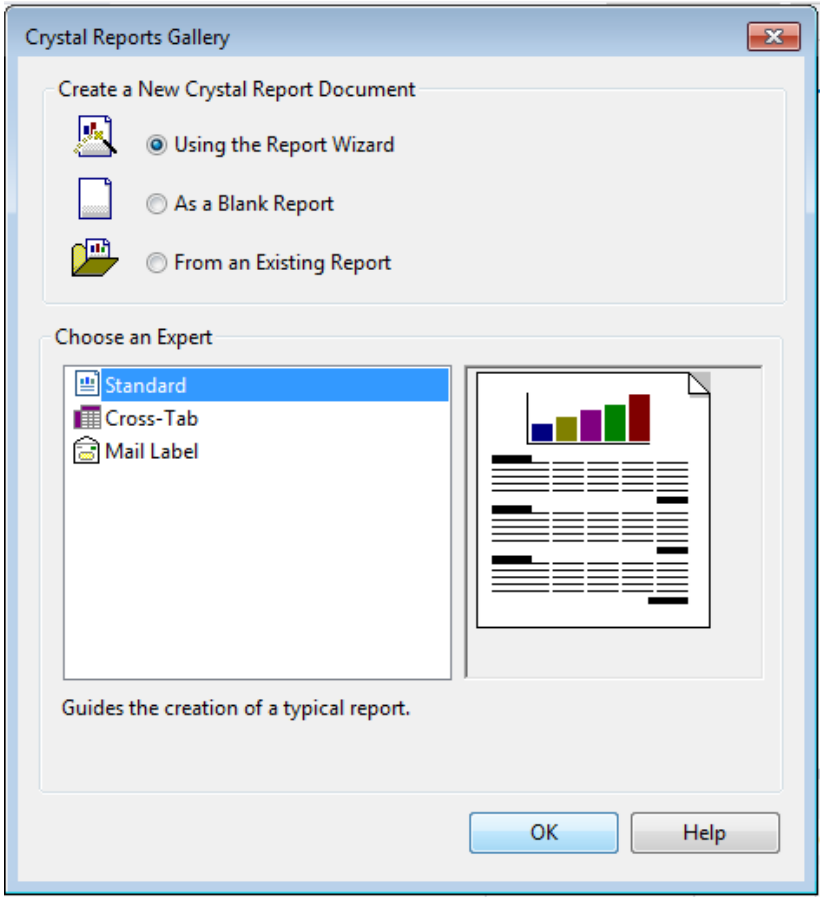
Crystal Reports Creation(cont'd)



Crystal Reports Creation(cont'd)



Crystal Reports Creation(cont'd)



Crystal Reports Creation(cont'd)

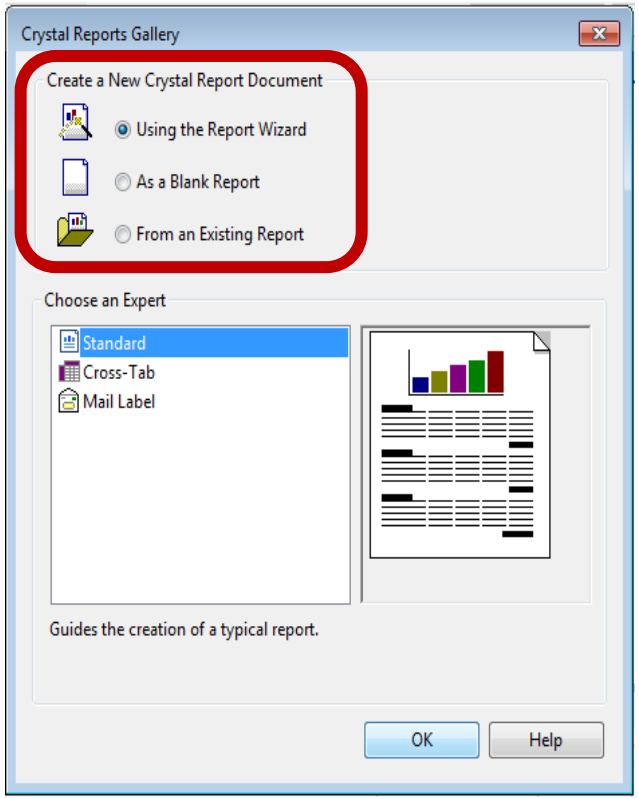
- Create a New Crystal Report Document
  - As a Blank Report

    Opens the Report Designer to build the report from scratch.
  - From an Existing Report

    Creates a new report with the same design as another report you specify.
  - Using the Report Wizard

    Create a new report using the report wizards, called Experts.

    These Experts walk you through a series of steps to select the data and format the report.



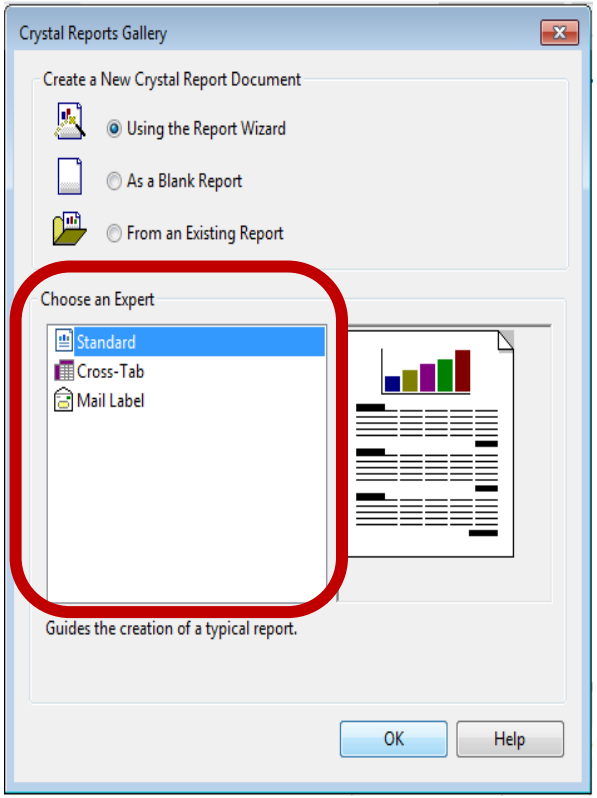
Crystal Reports Creation(cont'd)

- Choose an Expert (using the report wizard)
  - Standard Report

    Allows displaying lists of data and grouping. (typical report)
  - Cross-Tab Report

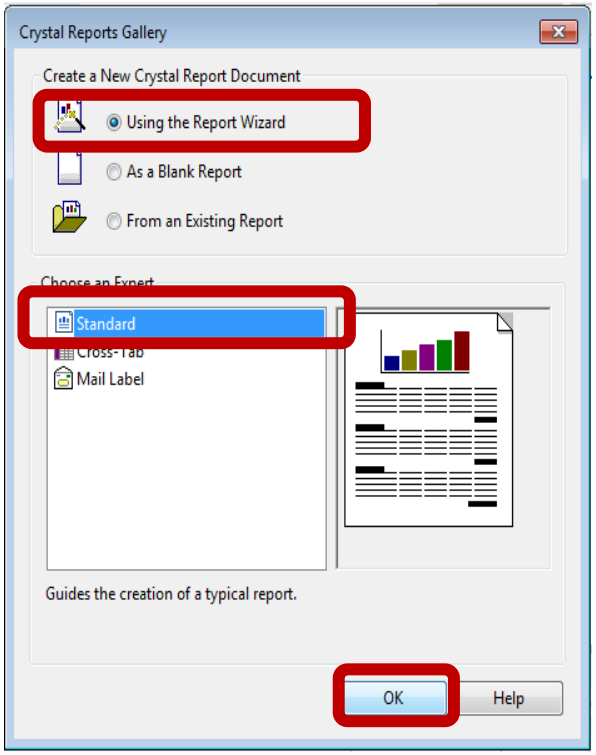
    Allows transforming data into grid (columns and rows).
  - Mail Label Report

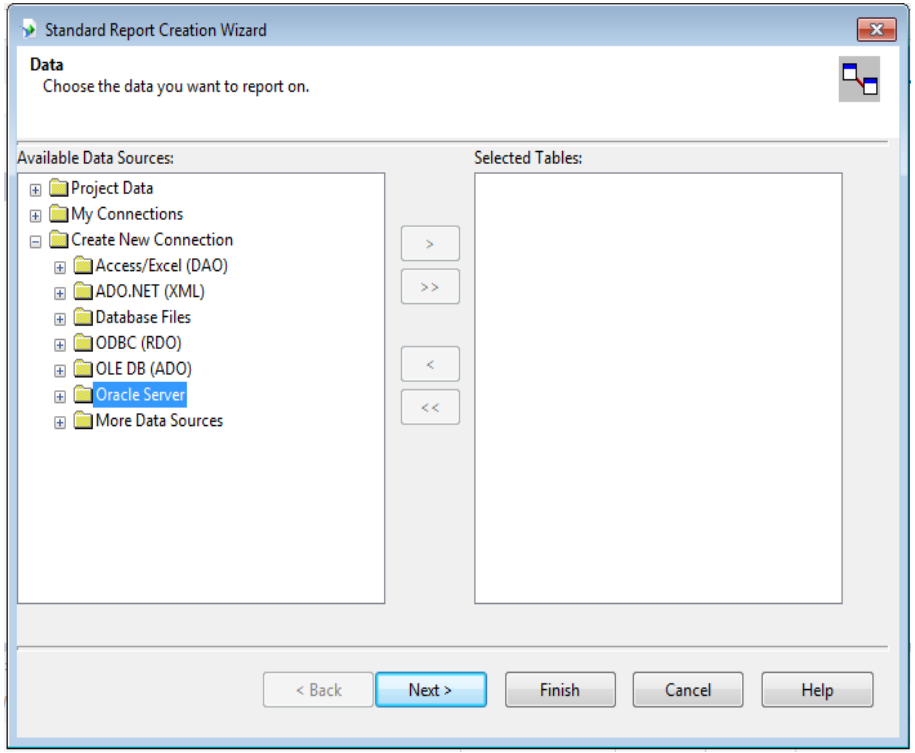
    Allows creating a report with multiple columns. (commonly used to create a sheet of address labels).



Crystal Reports Creation(cont'd)

Select Report Wizard and Standard Expert and then click on next.





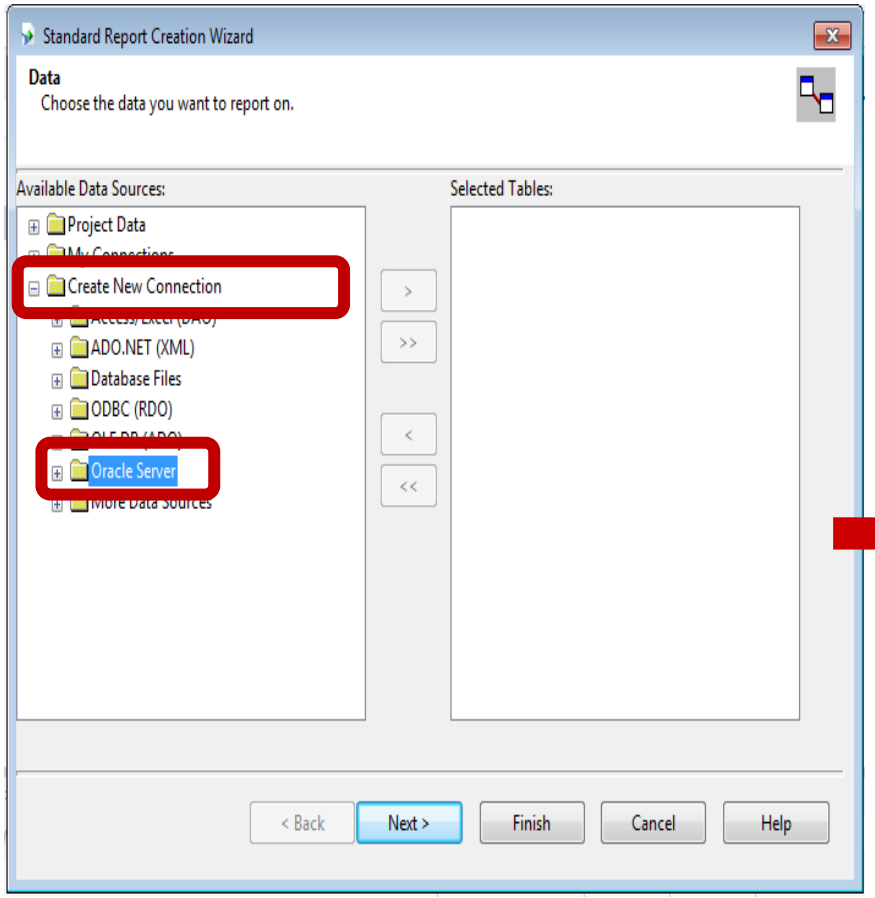
Crystal Reports Creation(cont'd)

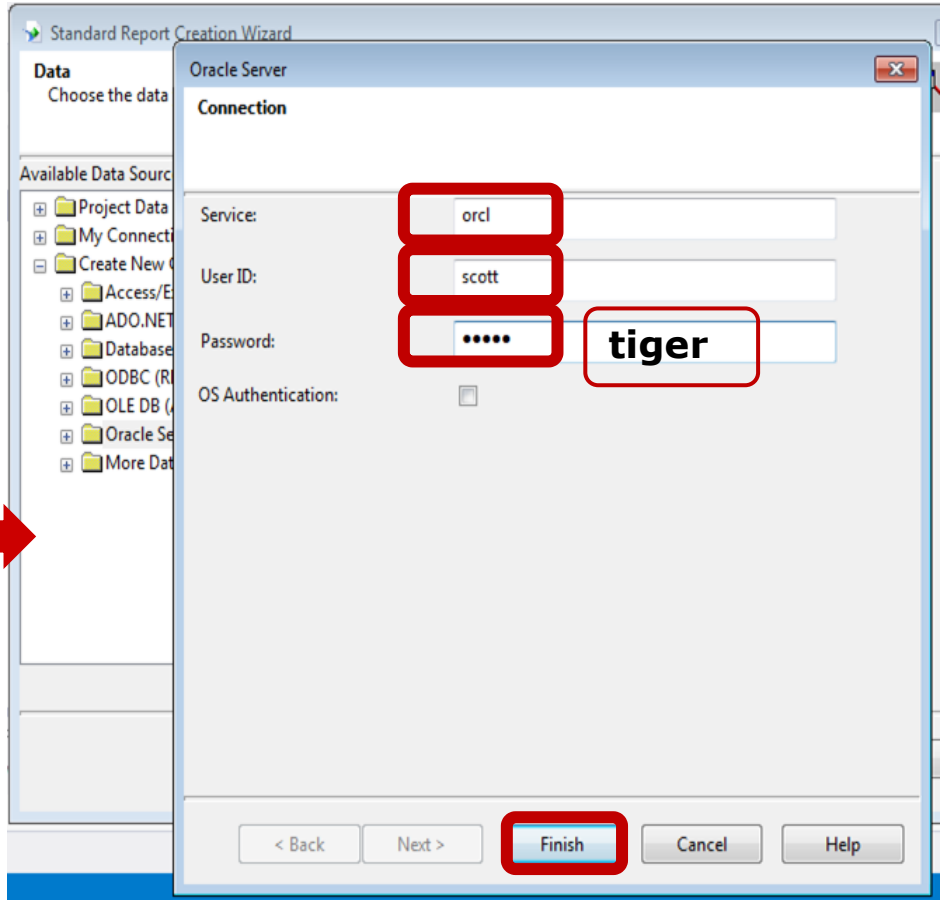
- Standard Report Creation wizard allows you to choose the data source that would be used in the report.
  - Create New Connection

    This option allows you to create a new connection to data source.
  - My Connections

    This Option lists the most recently used data source connection, so that you can quickly add it to your report.

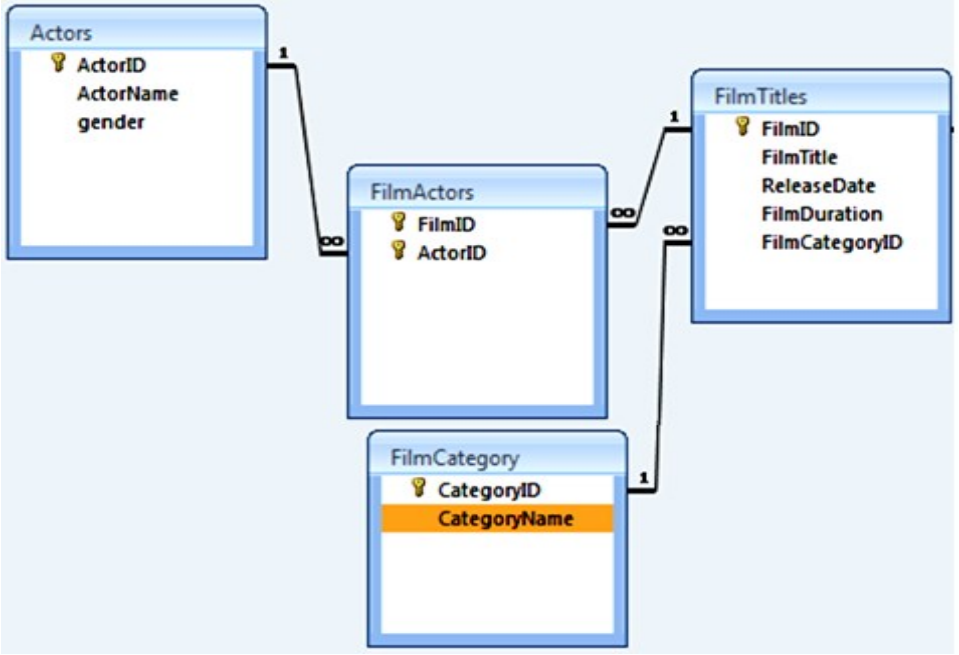
Crystal Reports Creation(cont'd)



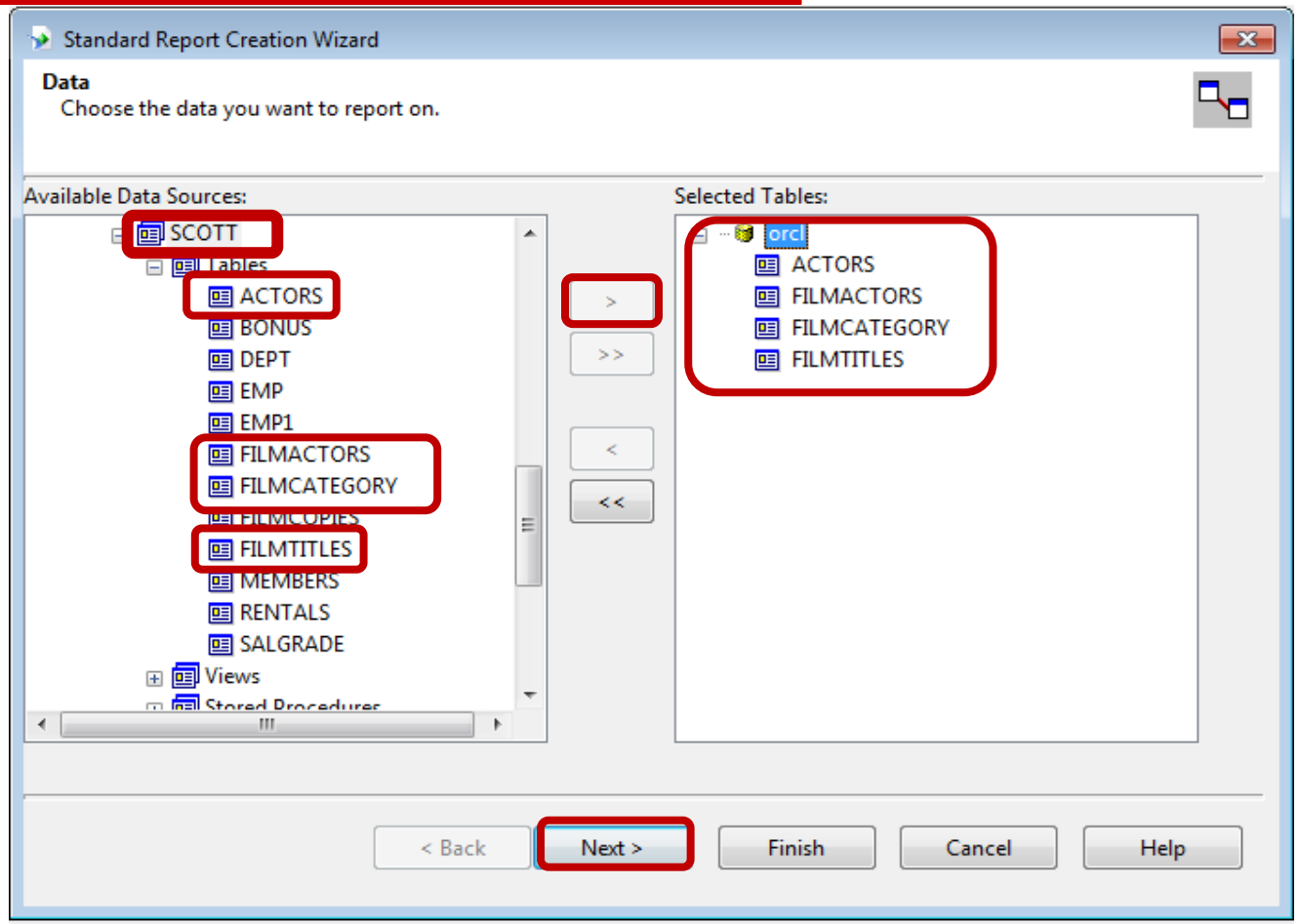


Crystal Reports Creation(cont'd)

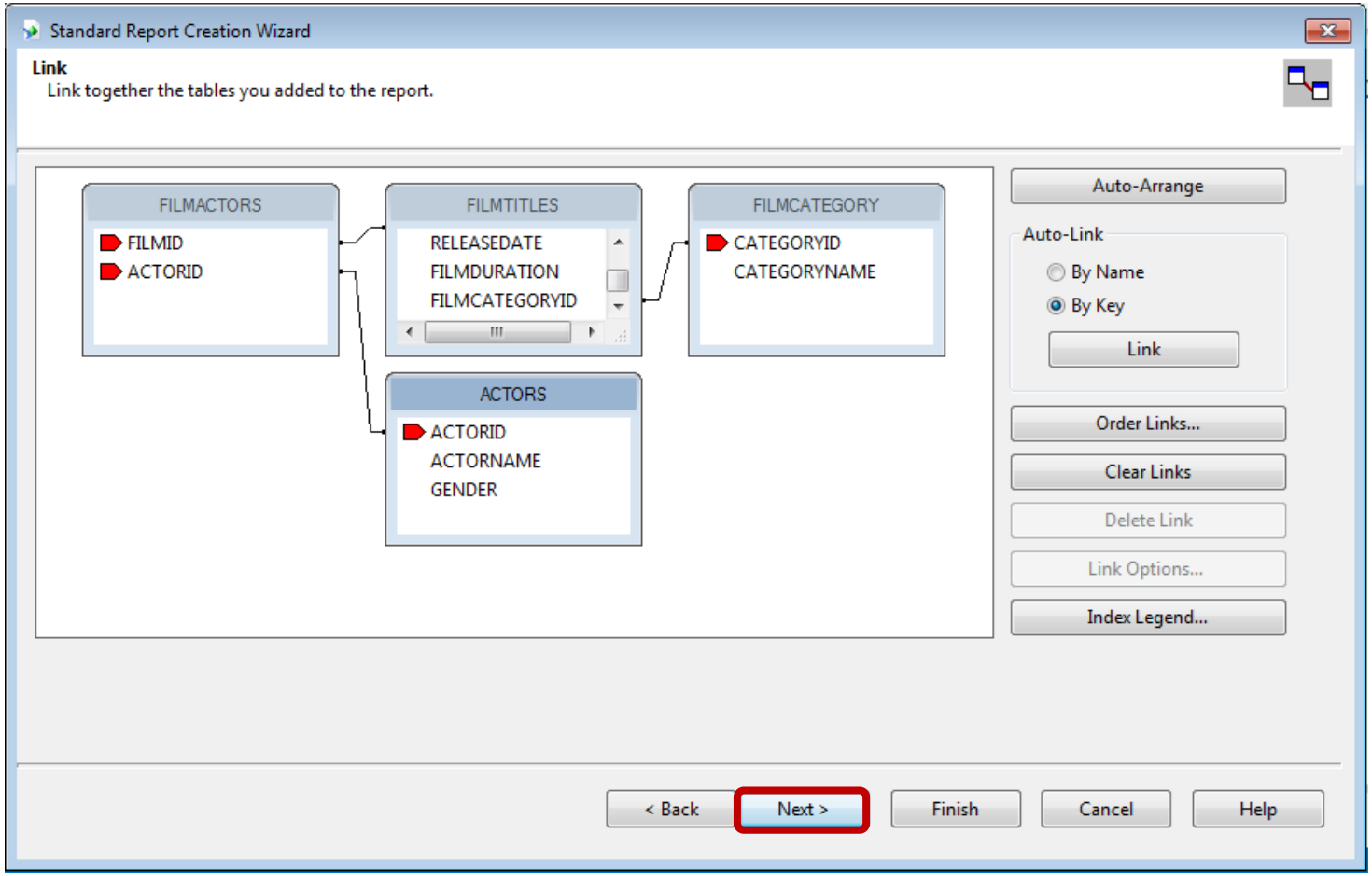
- Create a Report that displays the names of the movies for each category and the names of the actors for each movie title.



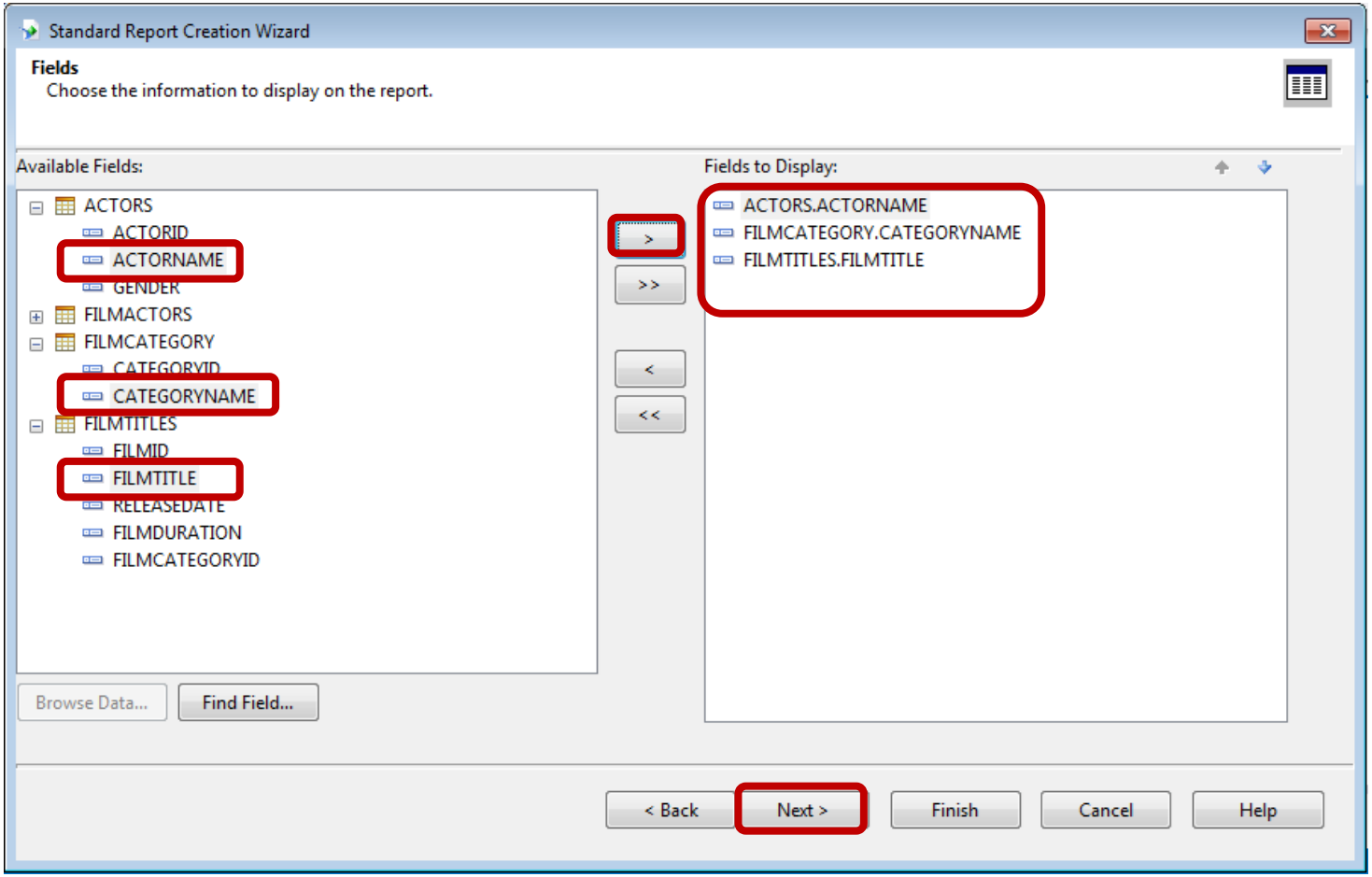
Crystal Reports Creation(cont'd)



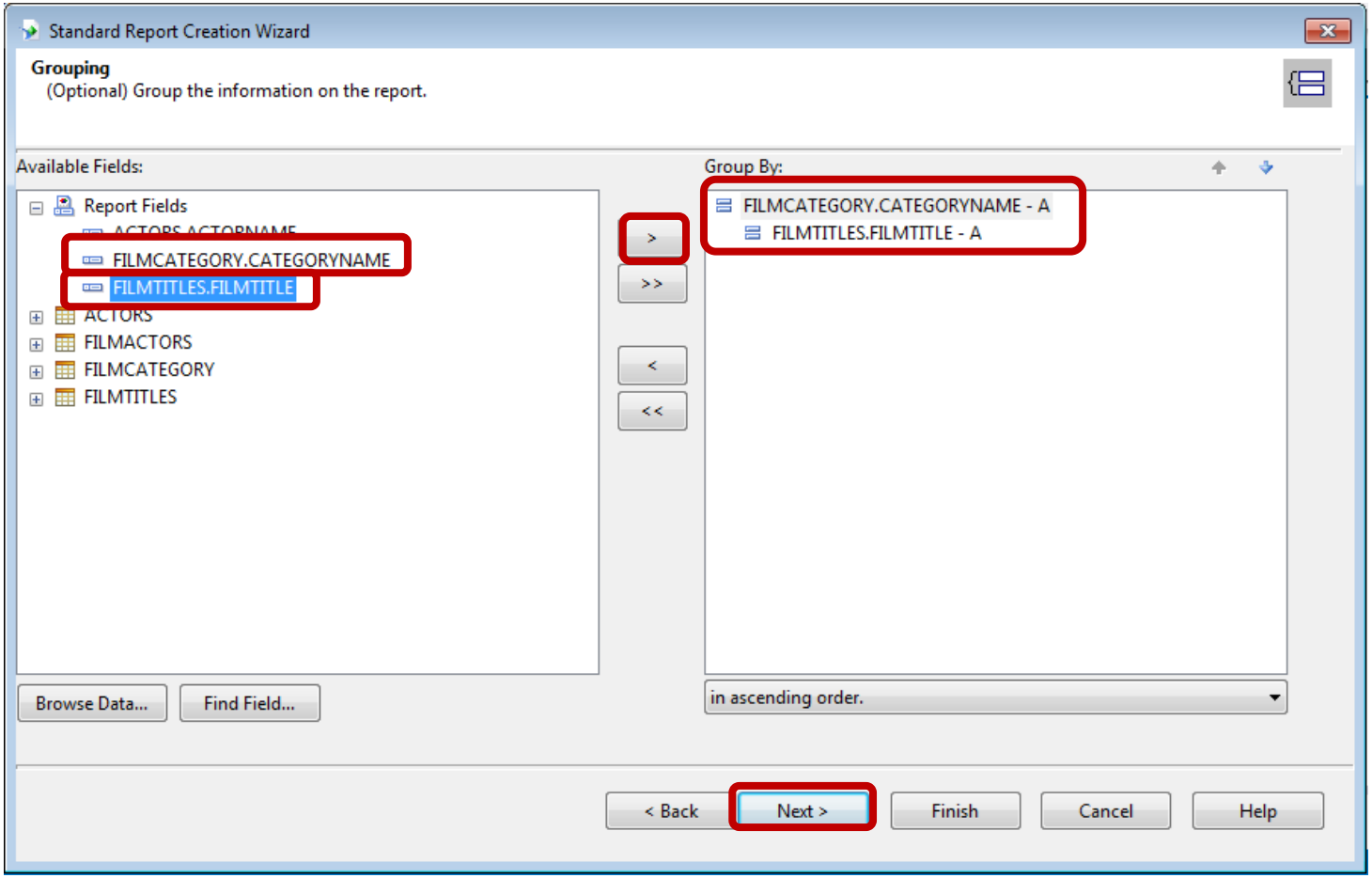
Crystal Reports Creation(cont'd)



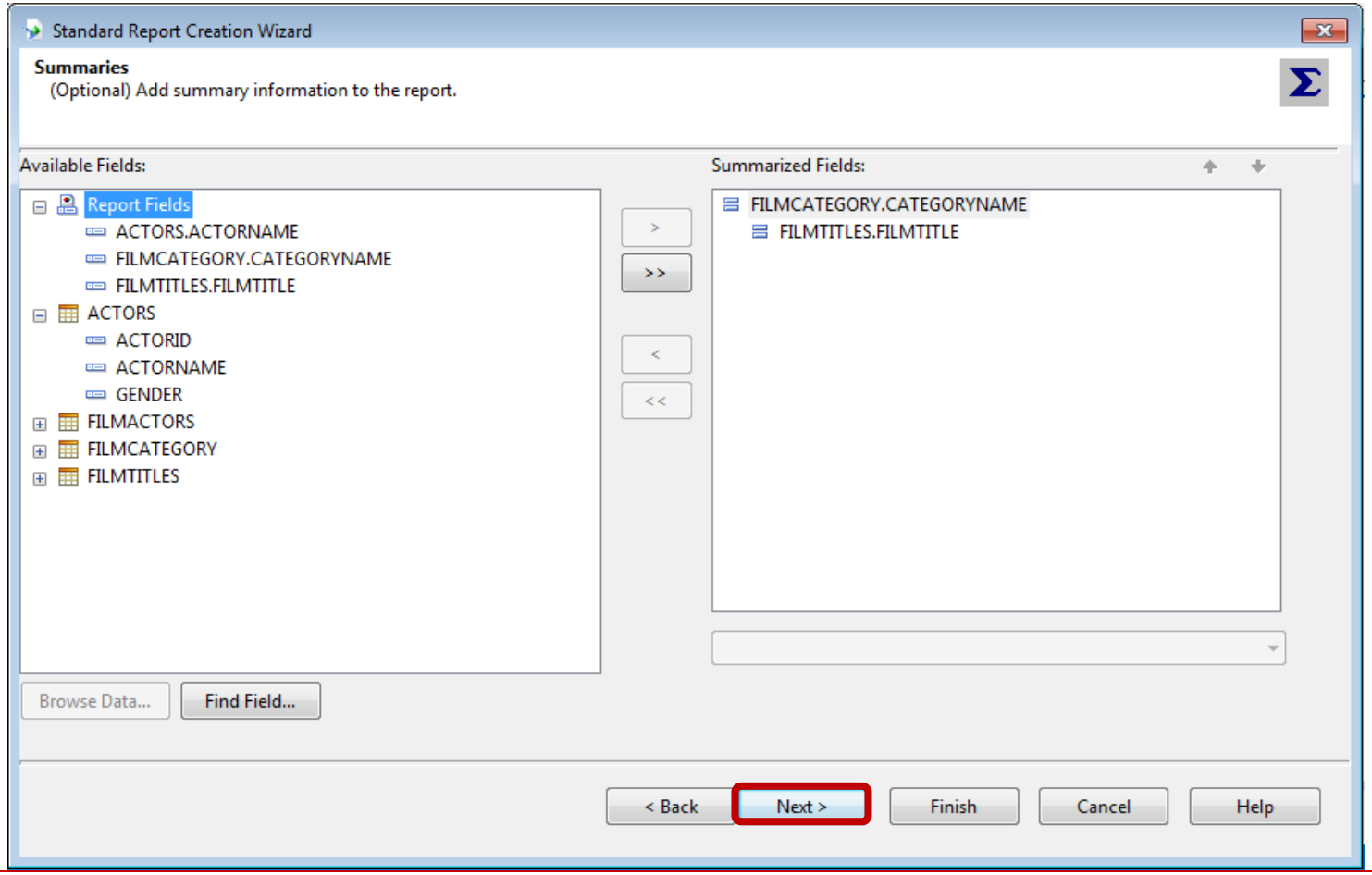
Crystal Reports Creation(cont'd)



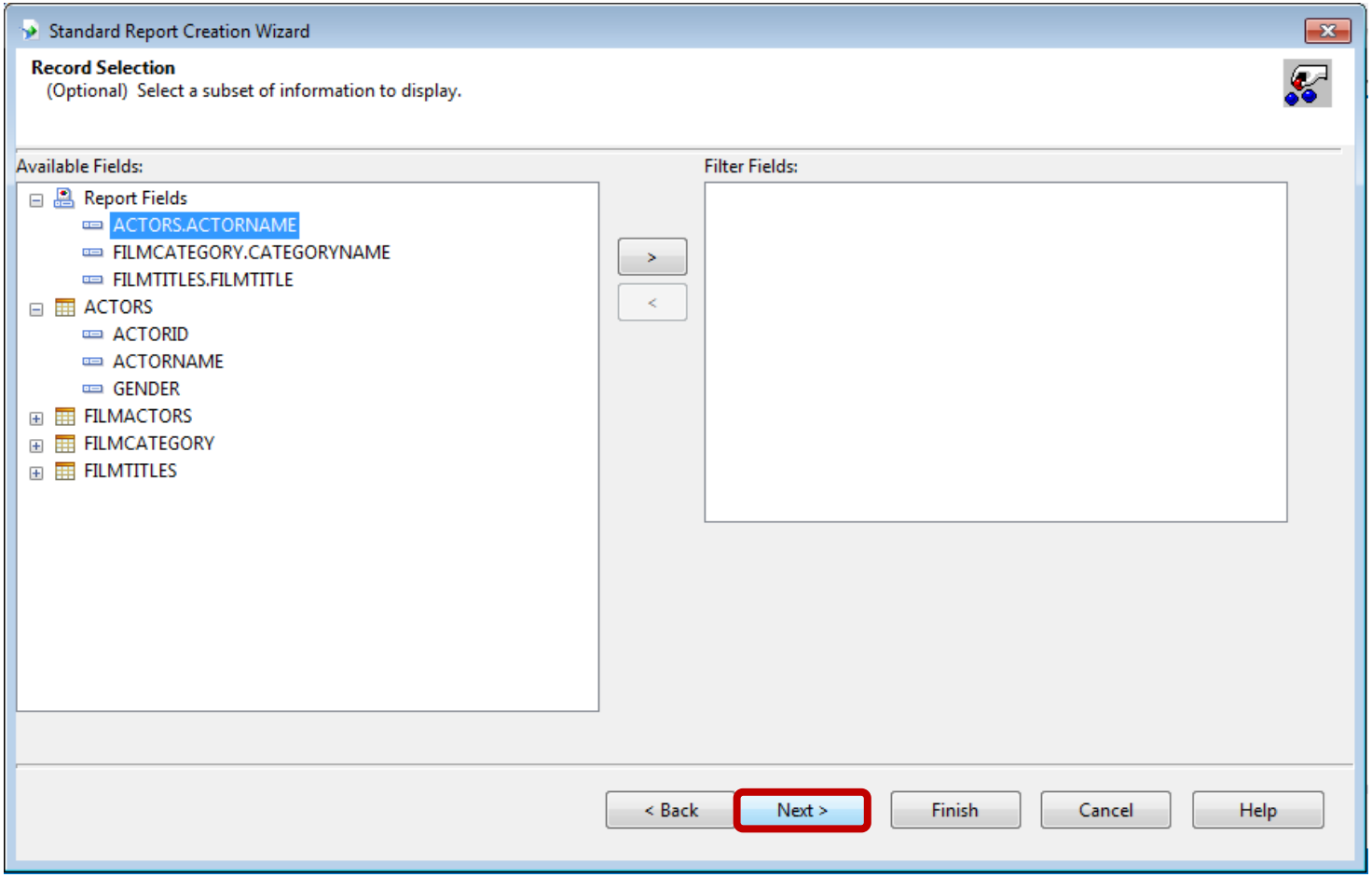
Crystal Reports Creation(cont'd)



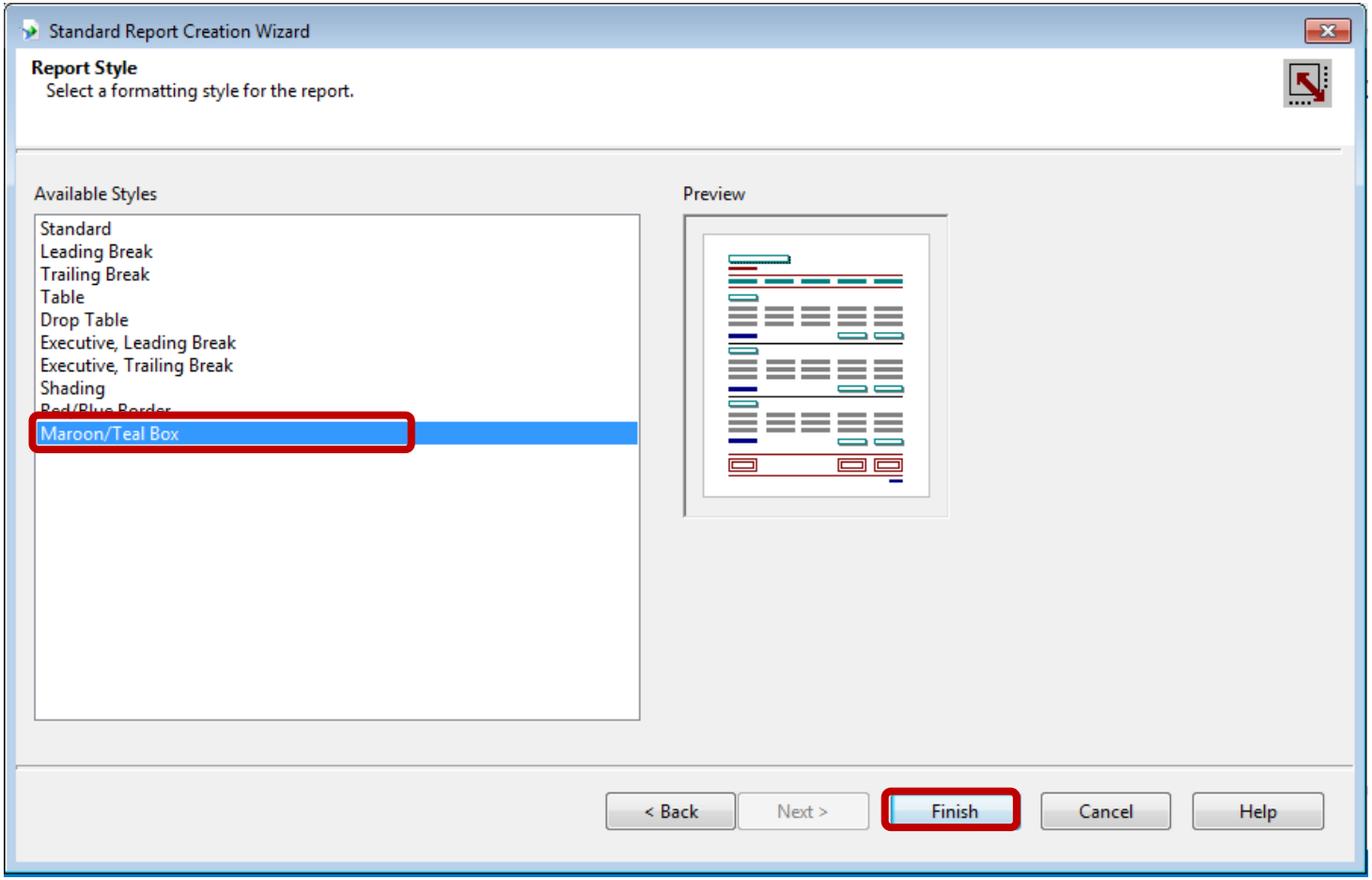
Crystal Reports Creation(cont'd)



Crystal Reports Creation(cont'd)



Crystal Reports Creation(cont'd)



Report Preview

FILE EDIT VIEW PROJECT BUILD DEBUG TEAM TOOLS CRYSTAL REPORTS WINDOW HELP

Start 100%

Docu... Toolbox CrystalReport1.rpt* Form1.cs [Design]

There are no items to show for the selected document.

Search Toolbox

Crystal Reports
- Pointer
- Text Object
- Line Object
- Box Object

General

There are no usable controls in this group. Drag an item onto this text to add it to the toolbox.

Save data in the report

1 / 1+ 85% SAP CRYSTAL REPORTS®

- Action
- Adventure
- Animation
- Comedy
- Fantasy
- Sci-Fi

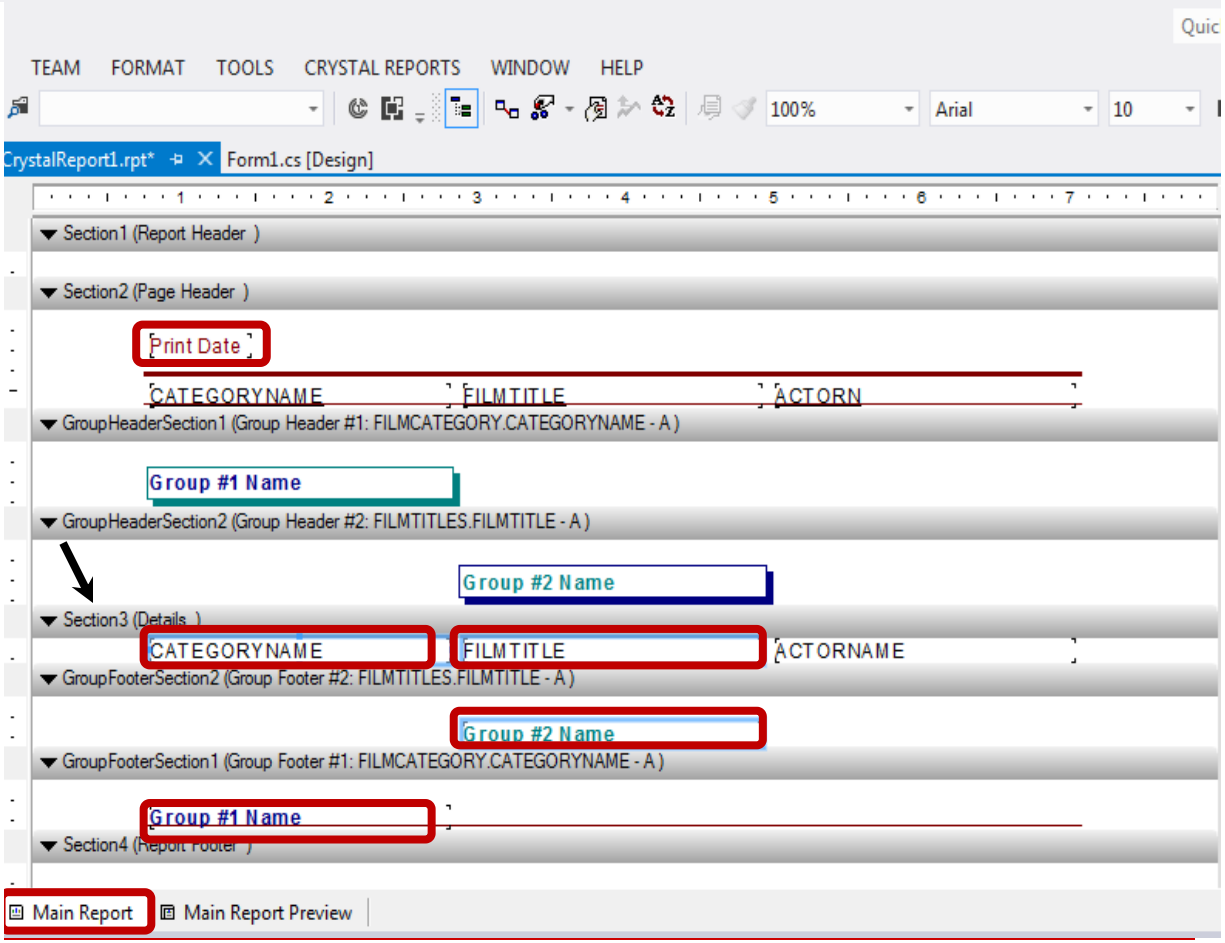
CATEGORY	FILMTITL	ACTORN
Action		
	Men in Black 3	
Action	Men in Black 3	George Clooney
Action	Men in Black 3	Will Smith
	Men in Black 3	
Action		
Adventure		
	The Twilight Saga Breaking Da	
Adventure	The Twilight Saga Breaking Dawı	Kristen Stewart
Adventure	The Twilight Saga Breaking Dawı	Robert Pattinson
Adventure	The Twilight Saga Breaking Dawı	Taylor Lautner
	The Twilight Saga Breaking Da	
	The Twilight Saga: Eclipse	
Adventure	The Twilight Saga: Eclipse	Kristen Stewart
Adventure	The Twilight Saga: Eclipse	Robert Pattinson
Adventure	The Twilight Saga: Eclipse	Taylor Lautner

Main Report | Main Report Preview

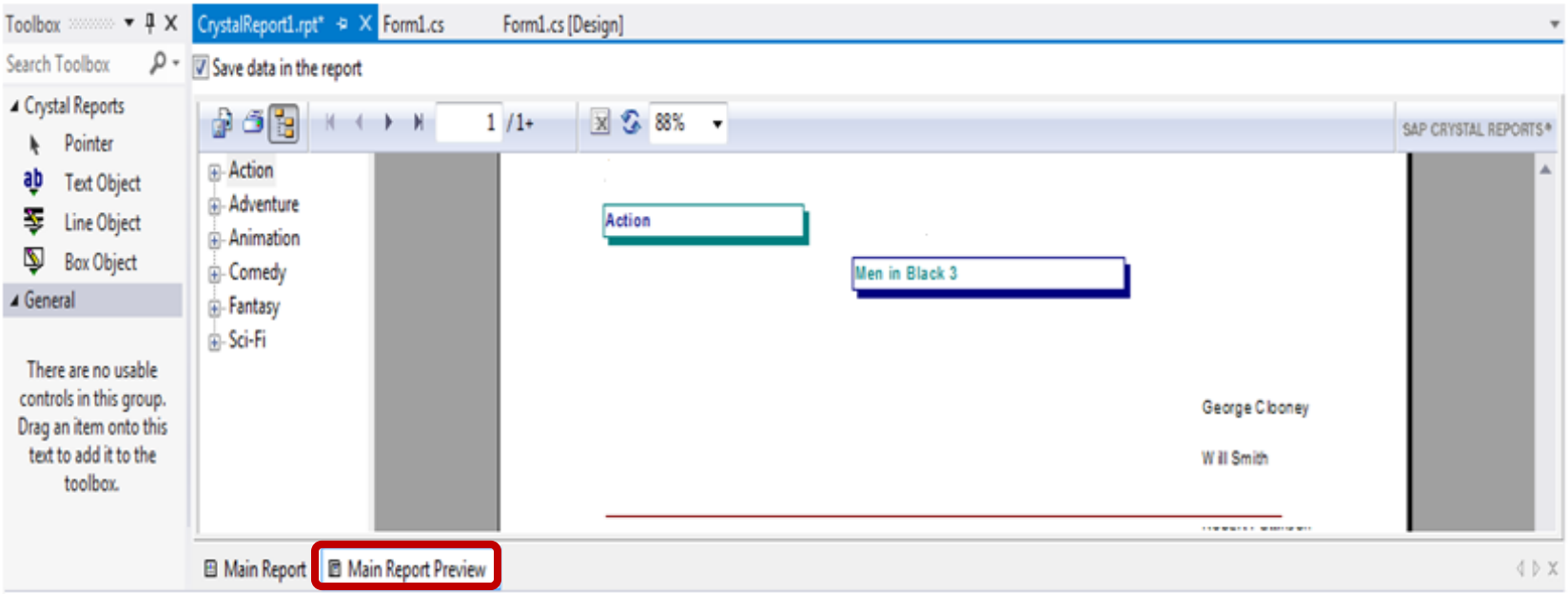
Modify Report Design

- Category name and film title are repeated for each actor name because they are included in the Details section which displays the included columns for each row

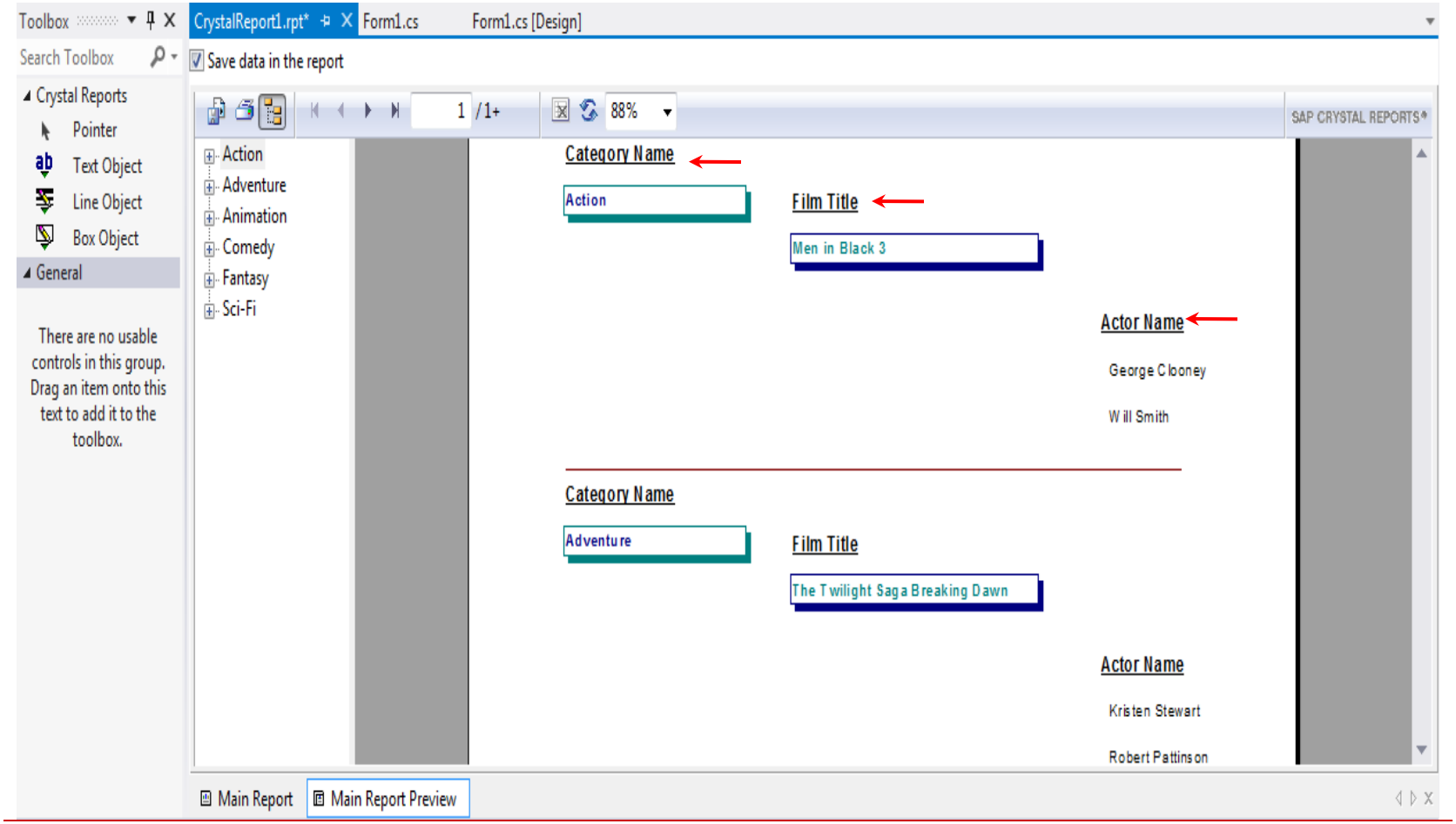
- To prevent repetition, select the column names and press the Del button



Modify Report Design(cont'd)

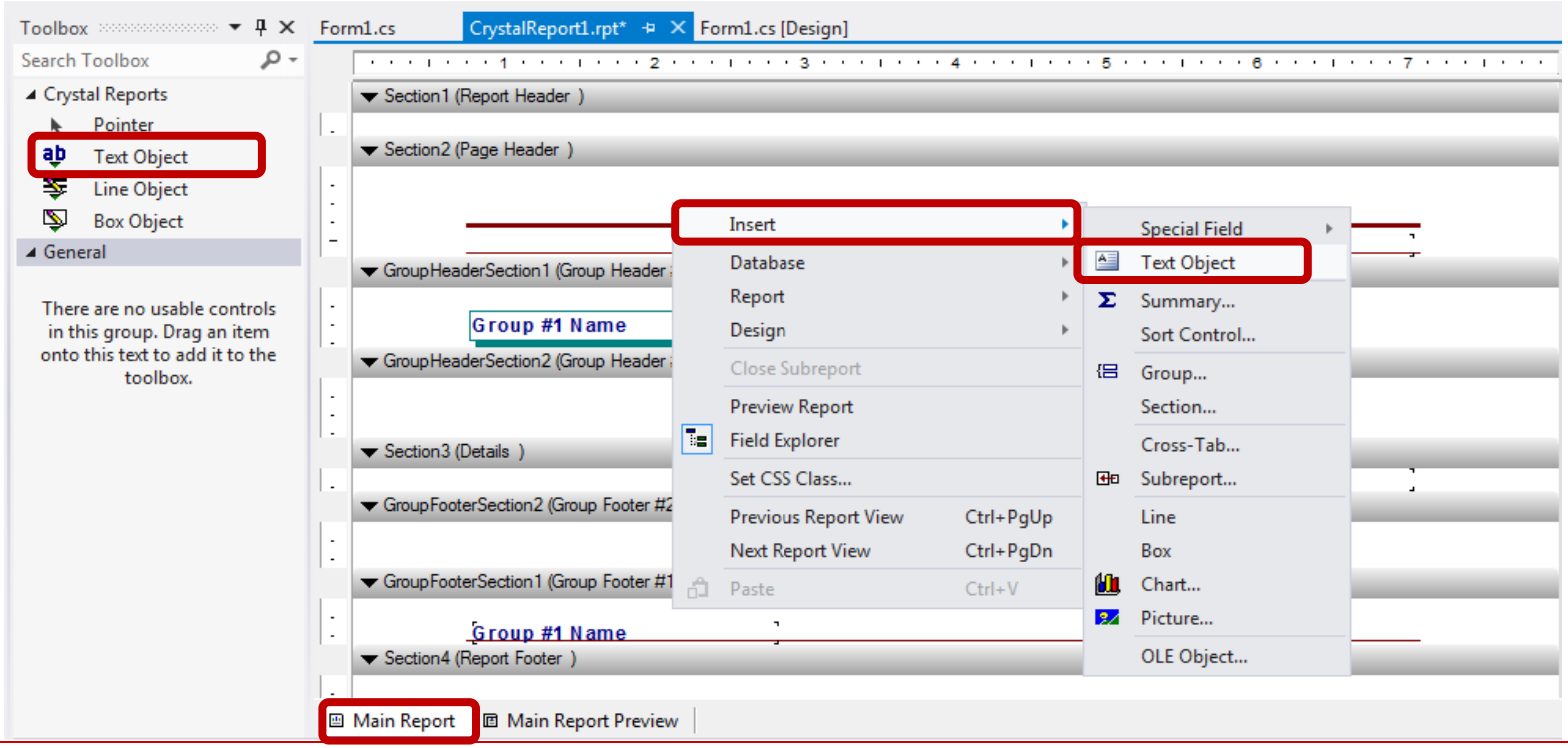


Inserting Labels into Report



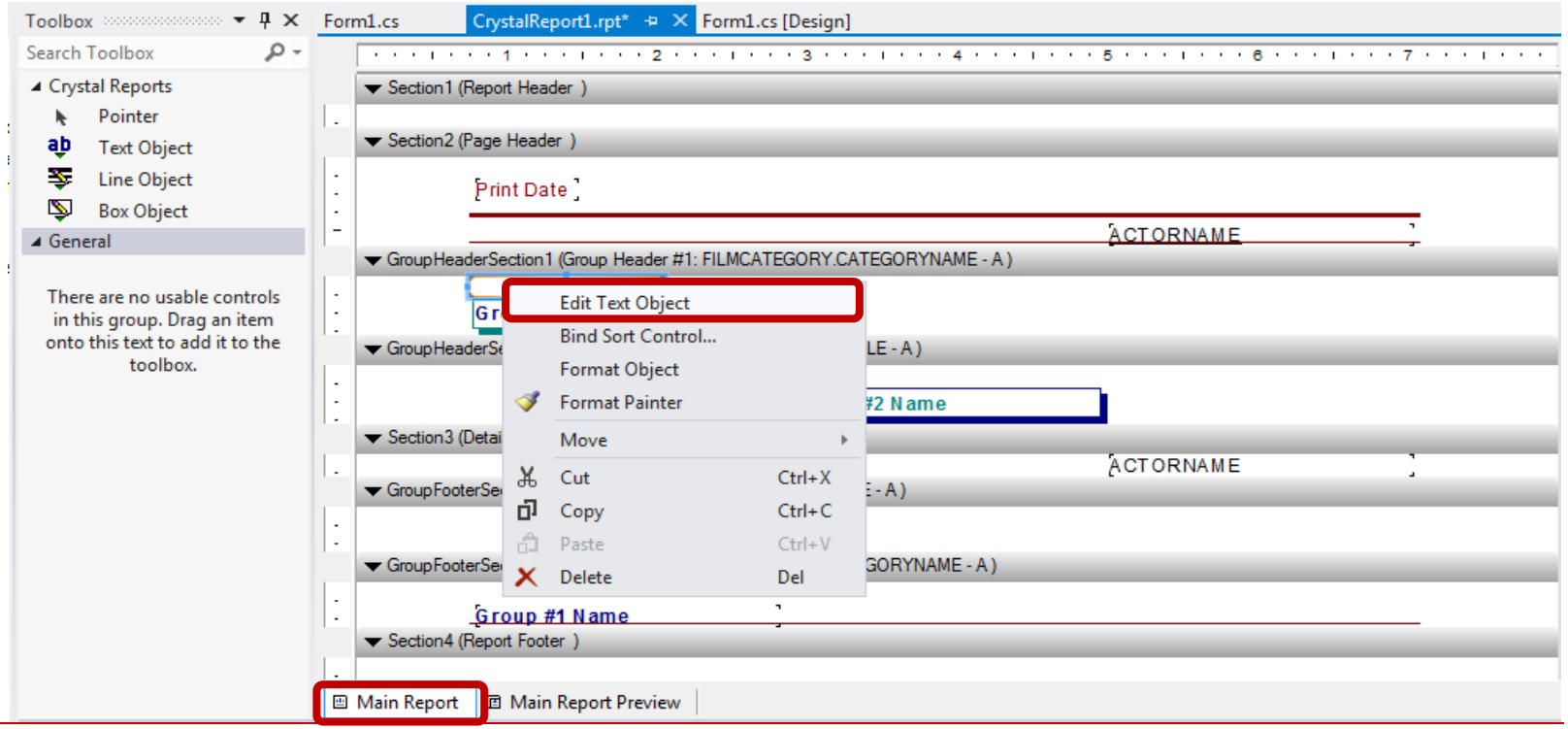
Inserting Labels into Report (cont'd)

- For Adding New Text Object, Right click on crystal reports designer window.



Inserting Labels into Report (cont'd)

- For Editing the text in the Text Object, Right click on Text Object then Edit Text Object.



Inserting Labels into Report (cont'd)

Toolbox
Search Toolbox
Crystal Reports
Pointer
Text Object
Line Object
Box Object
General
There are no usable controls in this group. Drag an item onto this text to add it to the toolbox.

Form1.cs | CrystalReport1.rpt | Form1.cs [Design]

GroupHeaderSection1 (Group Header #1: FILMCATEGORY.CATEGORYNAME - A)
Category Name
Film Title
Actor Name
Group #1 Name
GroupHeaderSection2 (Group H
Section3 (Details)
GroupFooterSection2 (Group Fo
GroupFooterSection1 (Group Fo
Section4 (Report Footer)
Main Report | Main Report P

Font
Font: Arial
Font style: Bold
Size: 11
Arial
Bookshelf Symbol 7
Calibri
Cambria
Cambria Math
Italic
Bold
Bold Italic
Black
Black Obliq
11 12 14 16 18 20 22
OK
Cancel
Effects
Strikeout
Underline
Sample
AaBbYyZz
Script:
Western

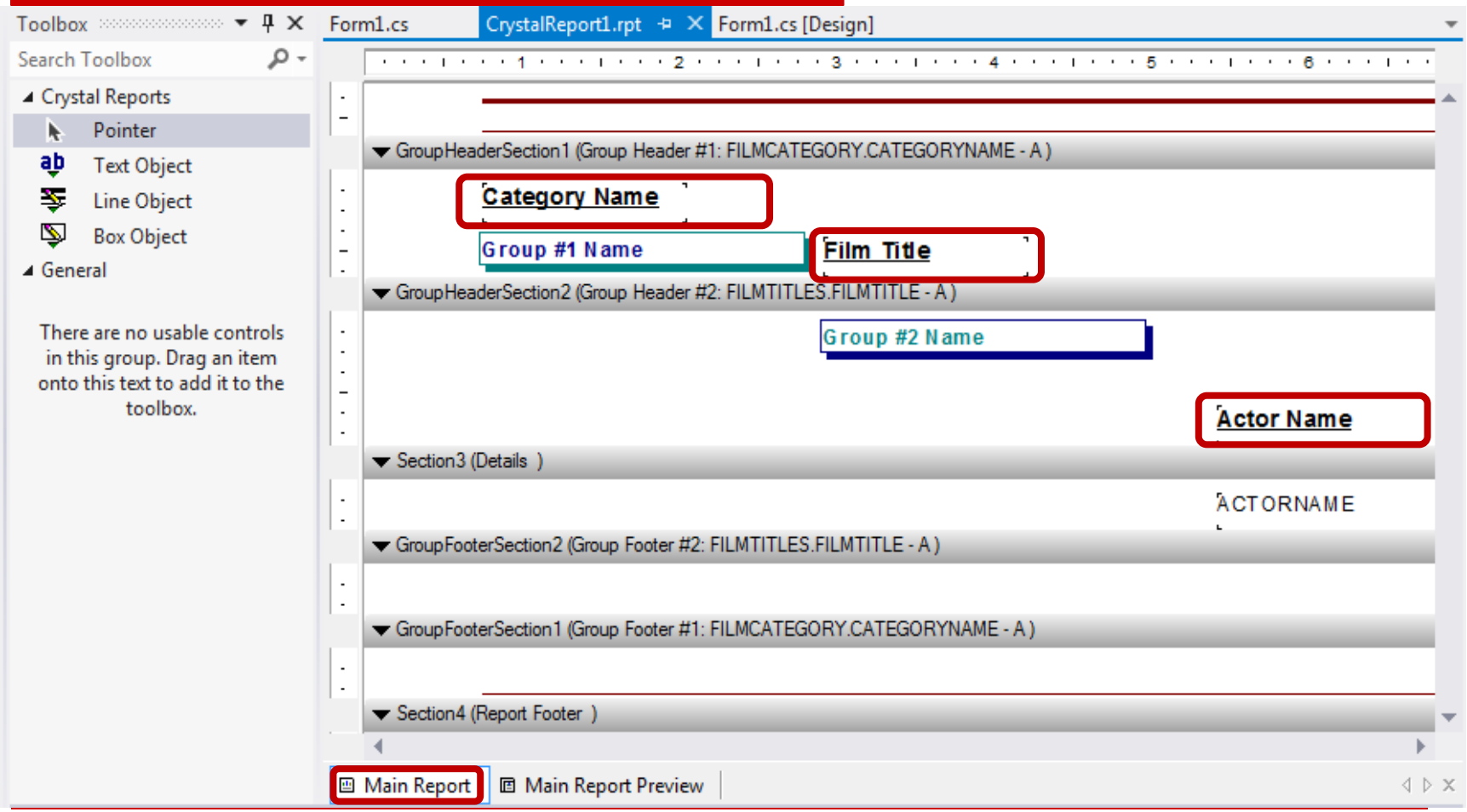
Solution Explorer
Search Solution Explorer (Ctrl+;)
CrystalReports_Lab1
Properties
References
App.config
CrystalReport1.rpt
Form1.cs

Properties
Text3 ITextObject

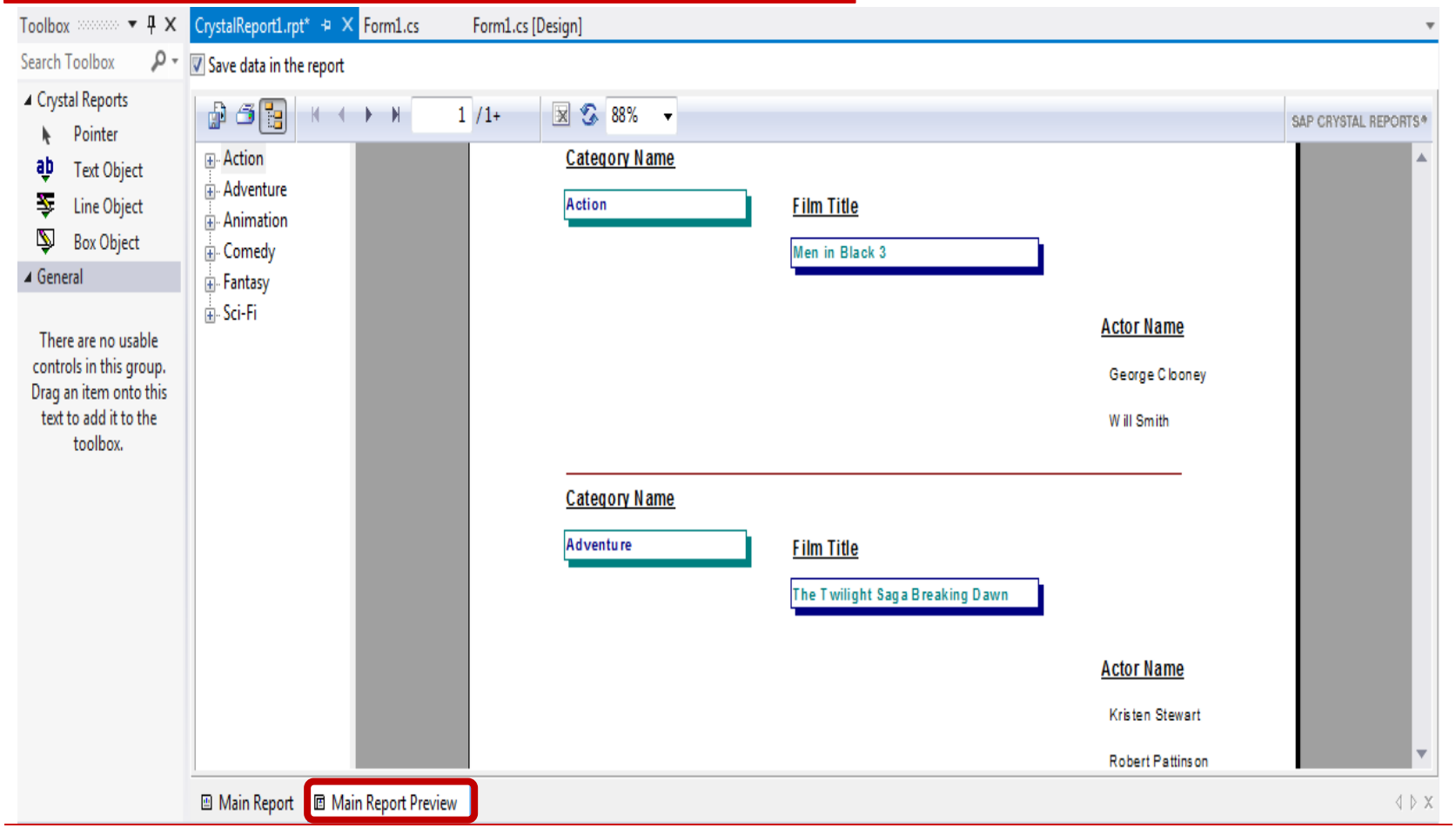
HorAlignment	crLeftAlign
KeepTogether	True
MaxNumberOfLines	0
Suppress	False
SuppressIfDuplicated	False
Font	
Font	Arial, 14.25pt, style=B
Bold	True
GdiCharSet	0
GdiVerticalFont	False

Font

Inserting Labels into Report (cont'd)

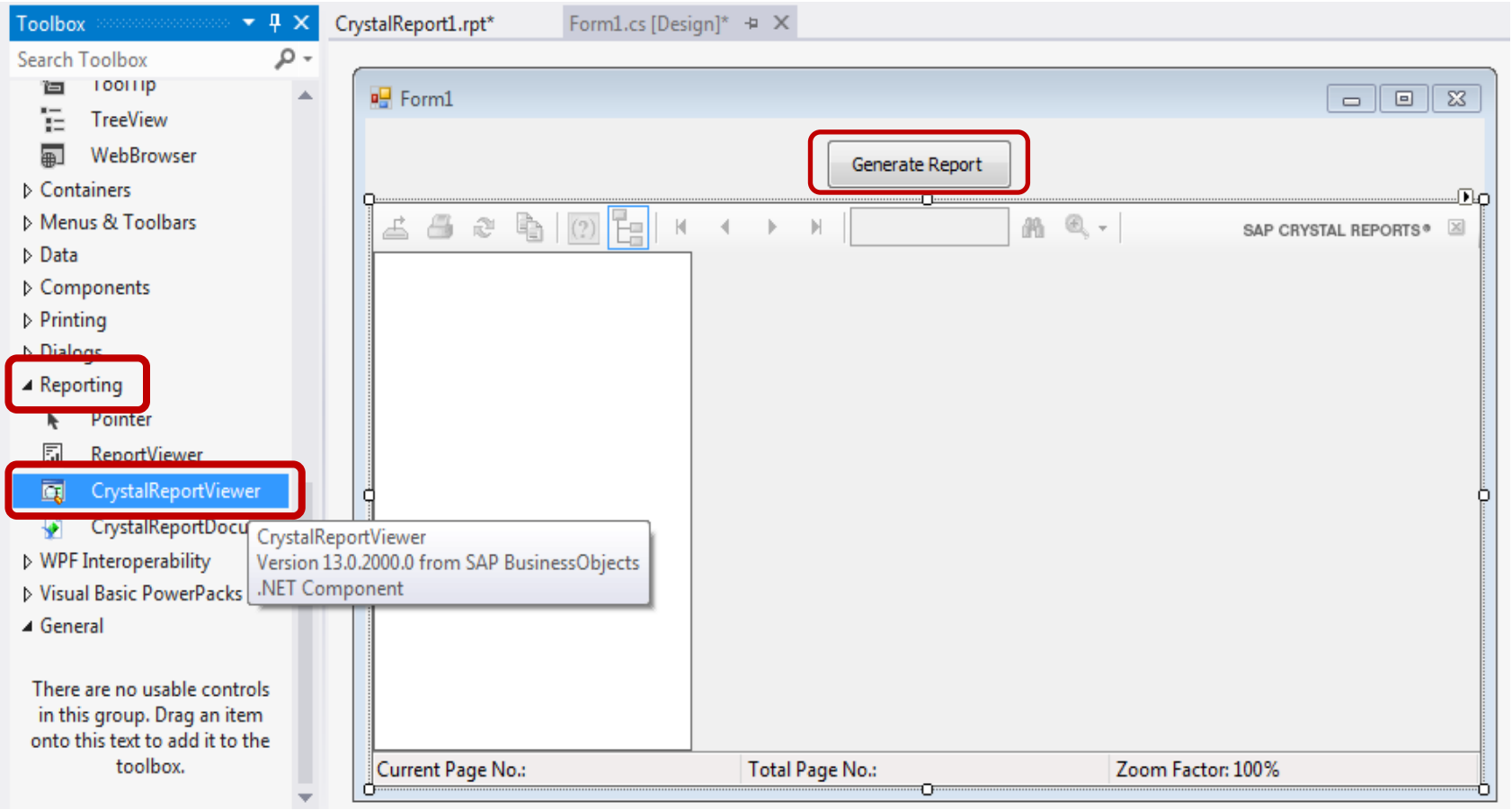


Inserting Labels into Report (cont'd)



Calling Crystal Reports From C# Form

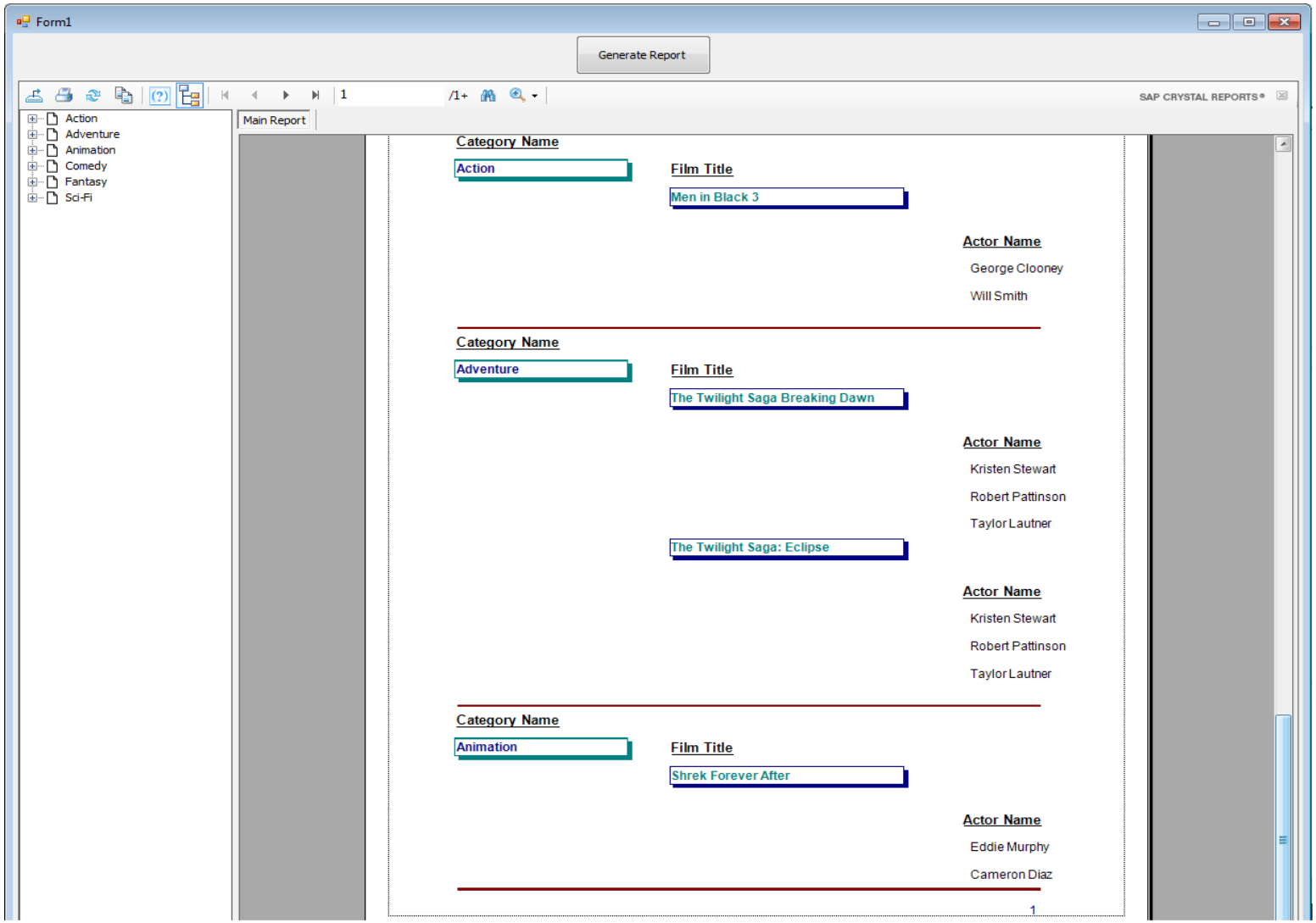
Crystal Reports Creation(cont'd)



Form1.cs* | Form1.cs [Design]* | Program.cs

CrystalReports_Lab1.Form1 | GenerateReport_btn_Click(object sender, EventArgs e)

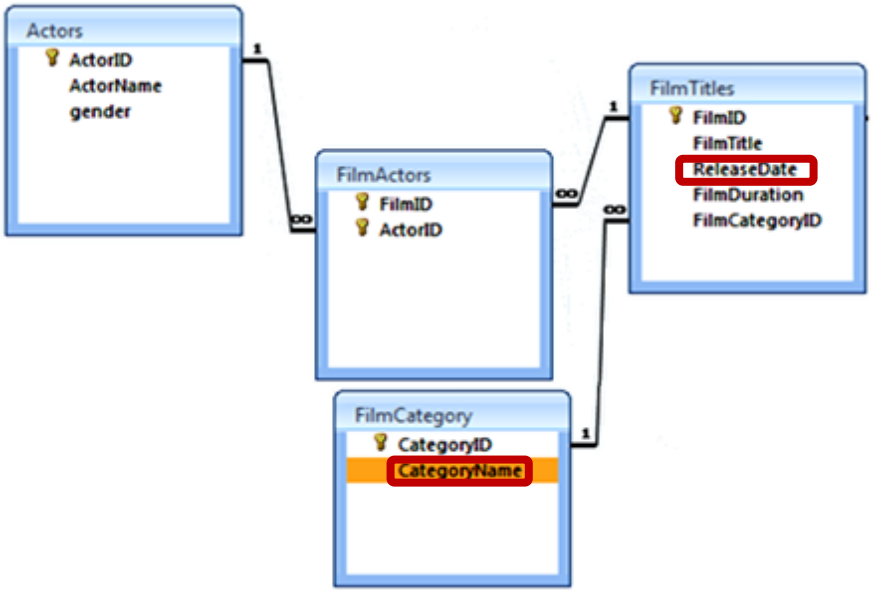
```
namespace CrystalReports_Lab1
{
    public partial class Form1 : Form
    {
        CrystalReport1 CR;

        public Form1()
        {
            InitializeComponent();
        }

        private void Form1_Load(object sender, EventArgs e)
        {
            CR = new CrystalReport1();
        }
        private void GenerateReport_btn_Click(object sender, EventArgs e)
        {
            crystalReportViewer1.ReportSource = CR;
        }
```


Form1
Generate Report
1
/1+
SAP CRYSTAL REPORTS
Action
Adventure
Animation
Comedy
Fantasy
Sci-Fi
Main Report
Category Name
Action
Film Title
Men in Black 3
Actor Name
George Clooney
Will Smith
Category Name
Adventure
Film Title
The Twilight Saga Breaking Dawn
Actor Name
Kristen Stewart
Robert Pattinson
Taylor Lautner
The Twilight Saga: Eclipse
Actor Name
Kristen Stewart
Robert Pattinson
Taylor Lautner
Category Name
Animation
Film Title
Shrek Forever After
Actor Name
Eddie Murphy
Cameron Diaz
1

Crystal Reports with Parameters

Crystal Reports with parameters

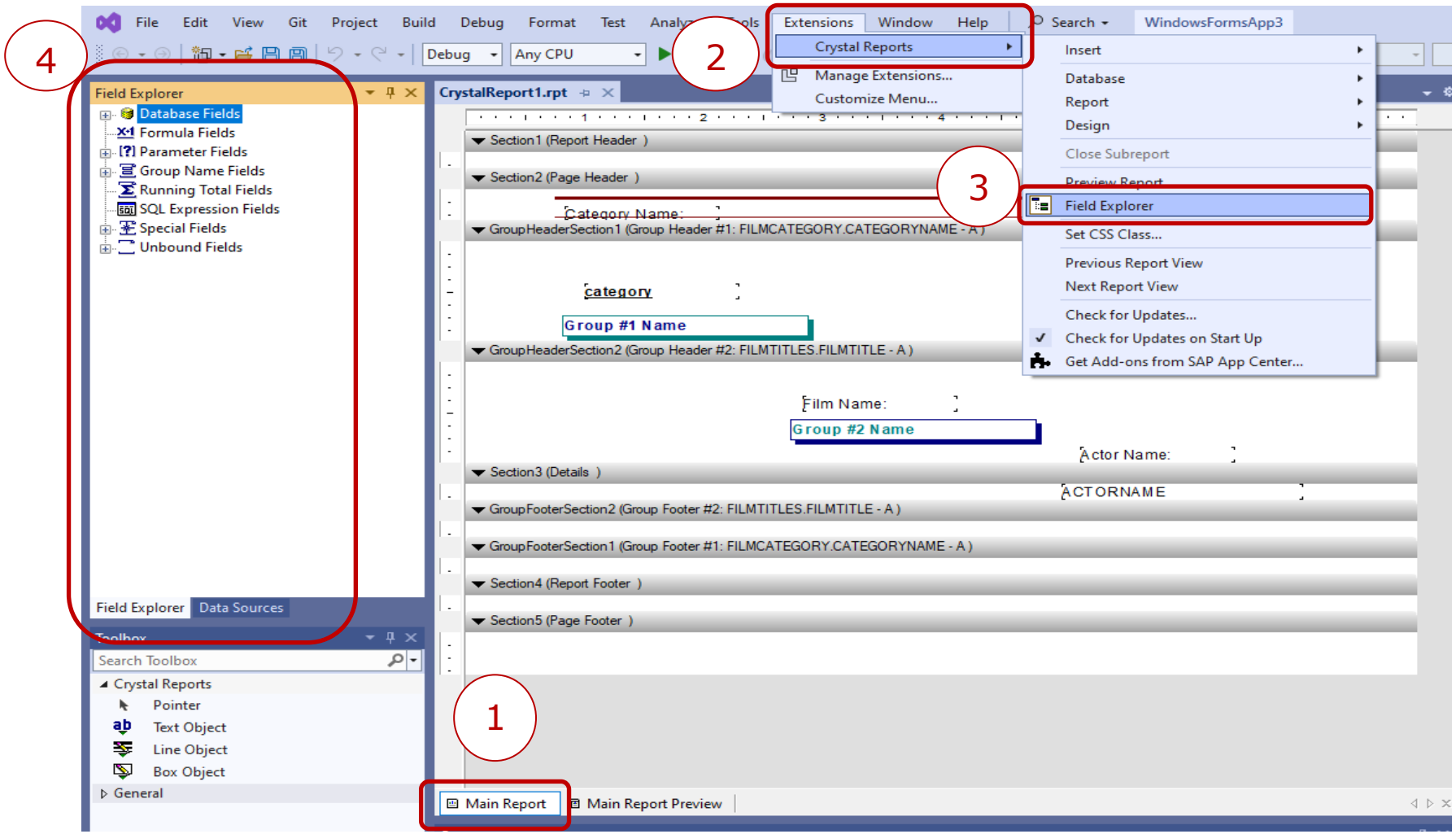
- Pass Category Name, Start Date, and End Date parameters to get a report that displays the names of all the movies for the selected category name whose Release Date is between the Start Date parameter and the End Date parameter and the names of the actors for these movies.



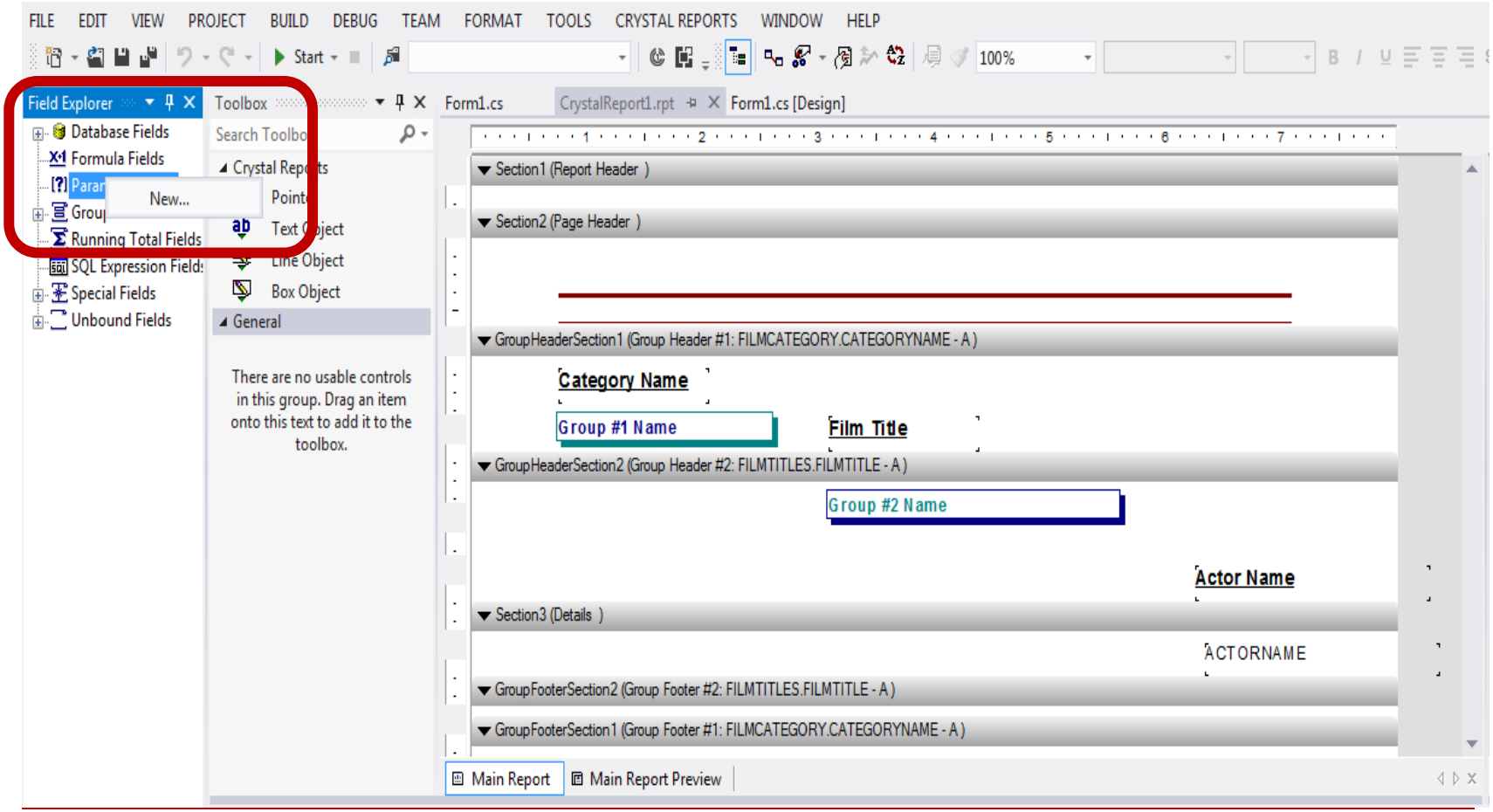
Crystal Reports with parameters(cont'd)

- To create a new parameter in Crystal report designer window.
  - Open the Main Report tab then Select the Field Explorer from Crystal Report Menu.

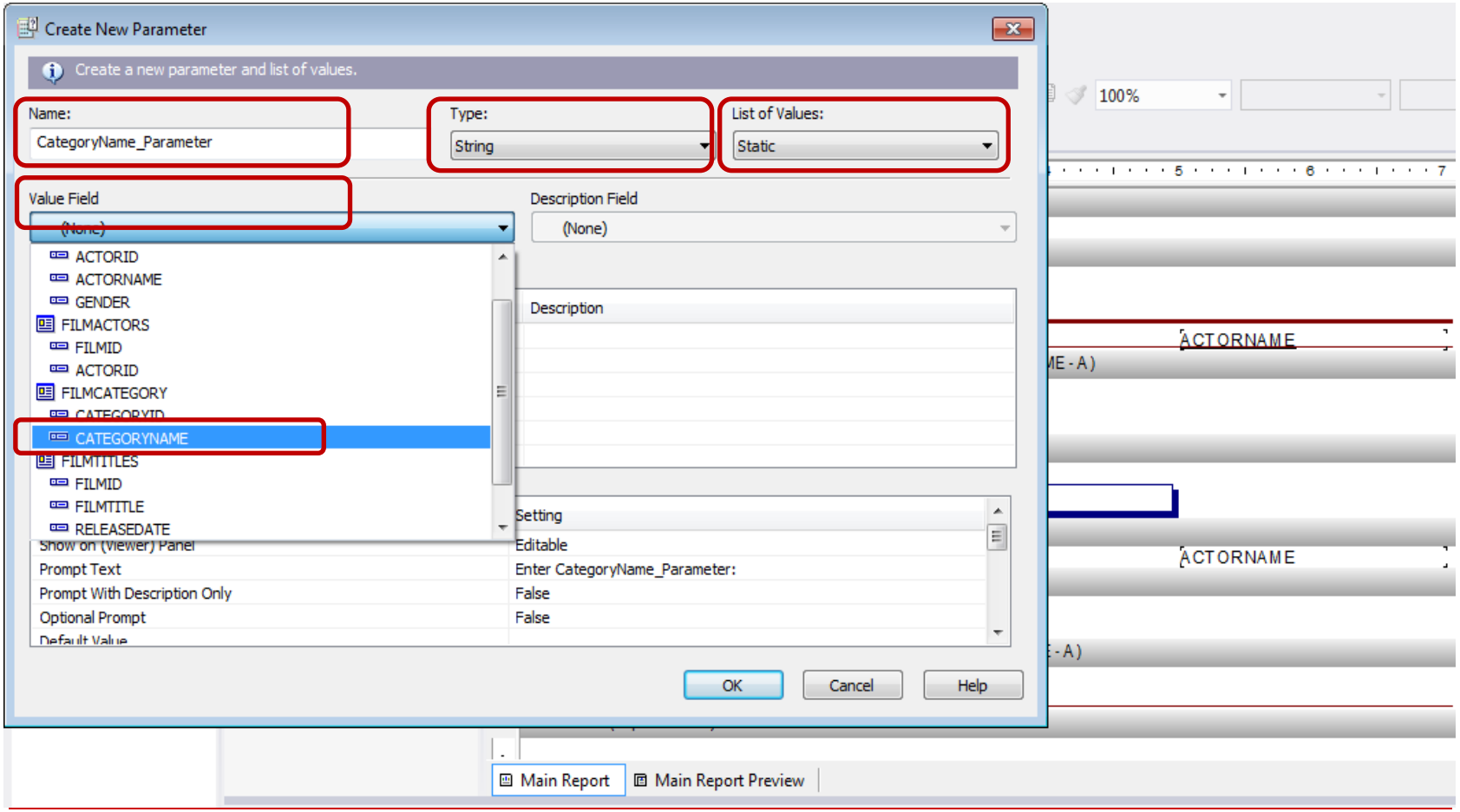
Crystal Reports with parameters(cont'd)



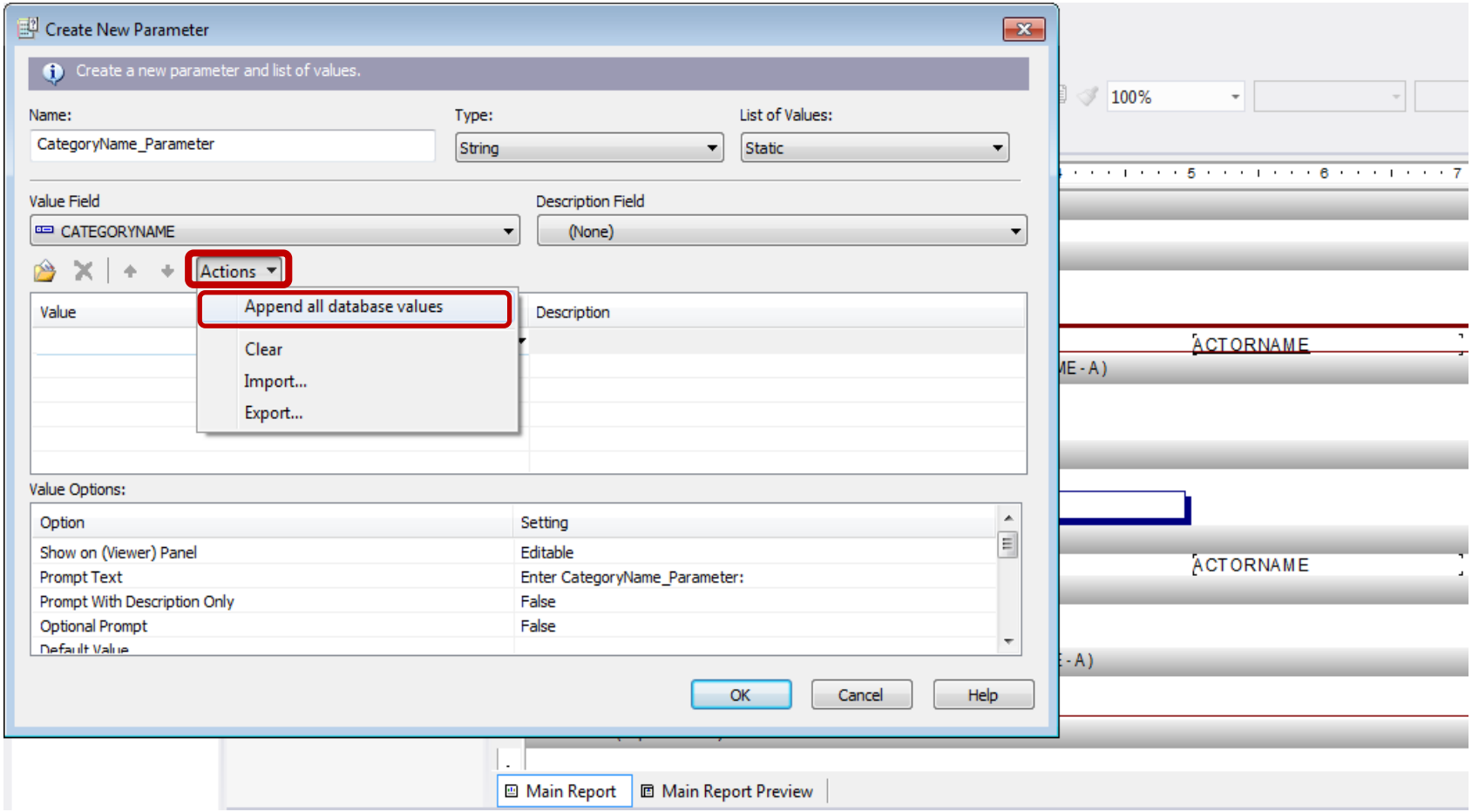
Crystal Reports with parameters(cont'd)



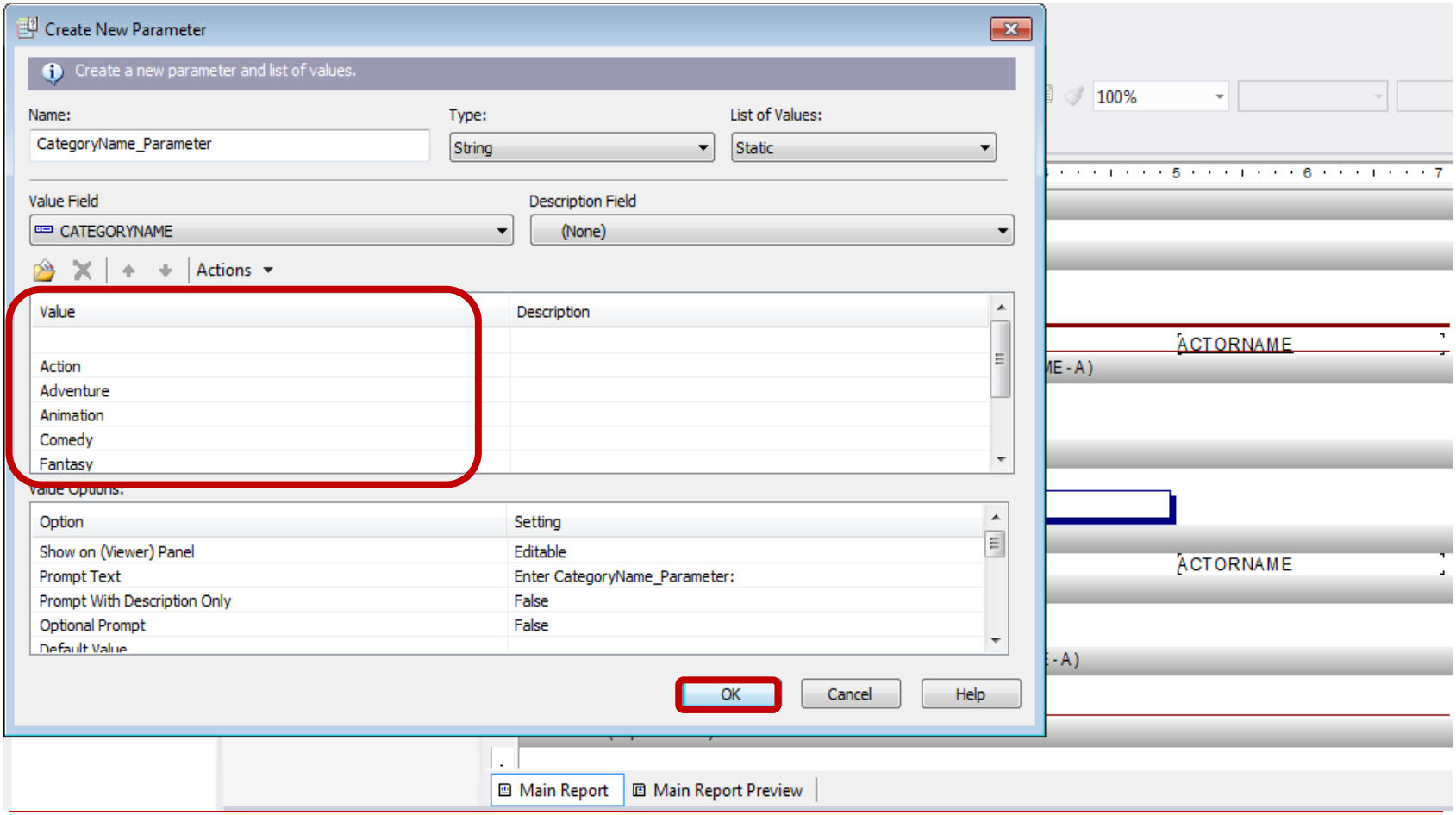
Crystal Reports with parameters(cont'd)



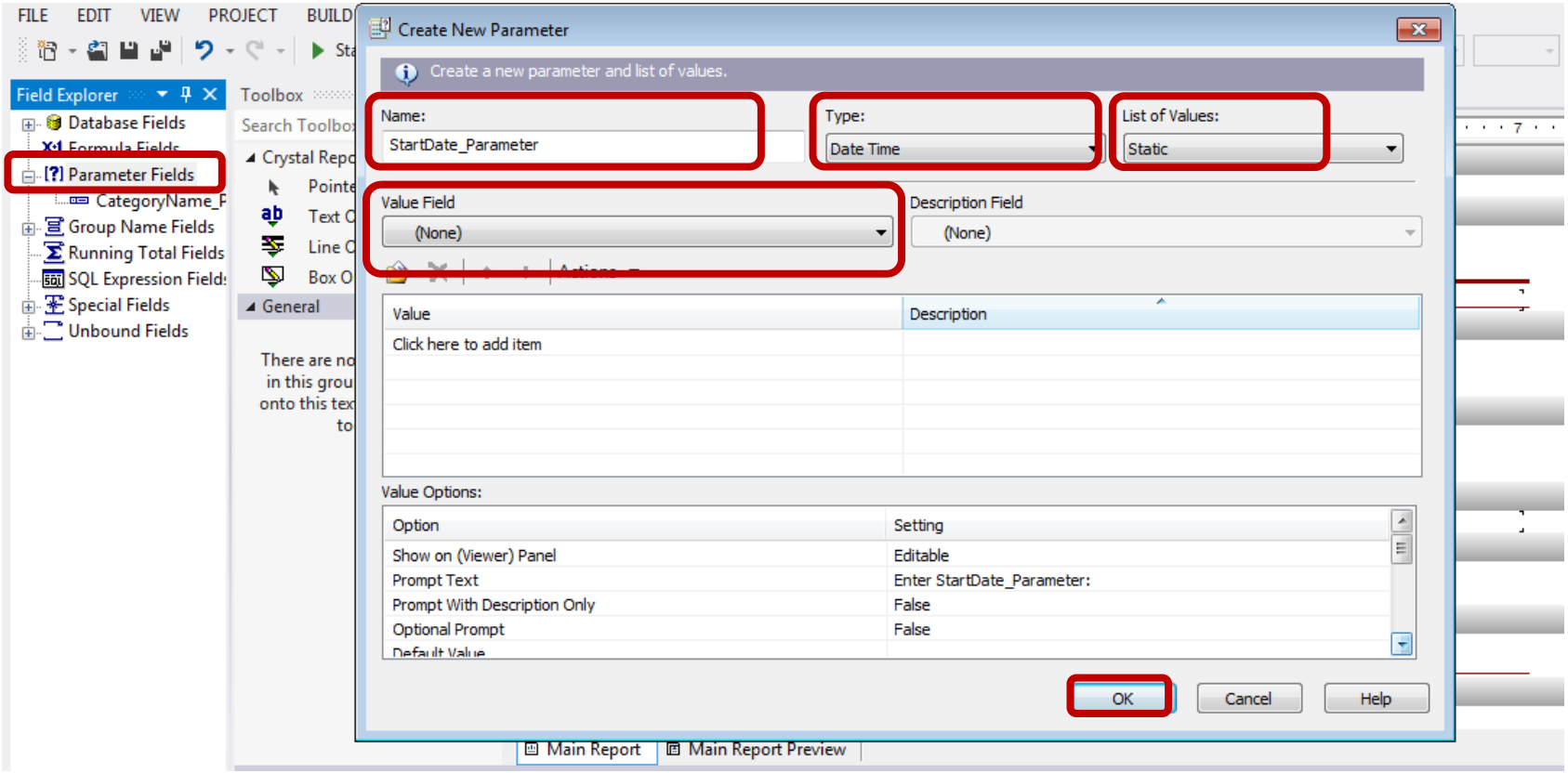
Crystal Reports with parameters(cont'd)



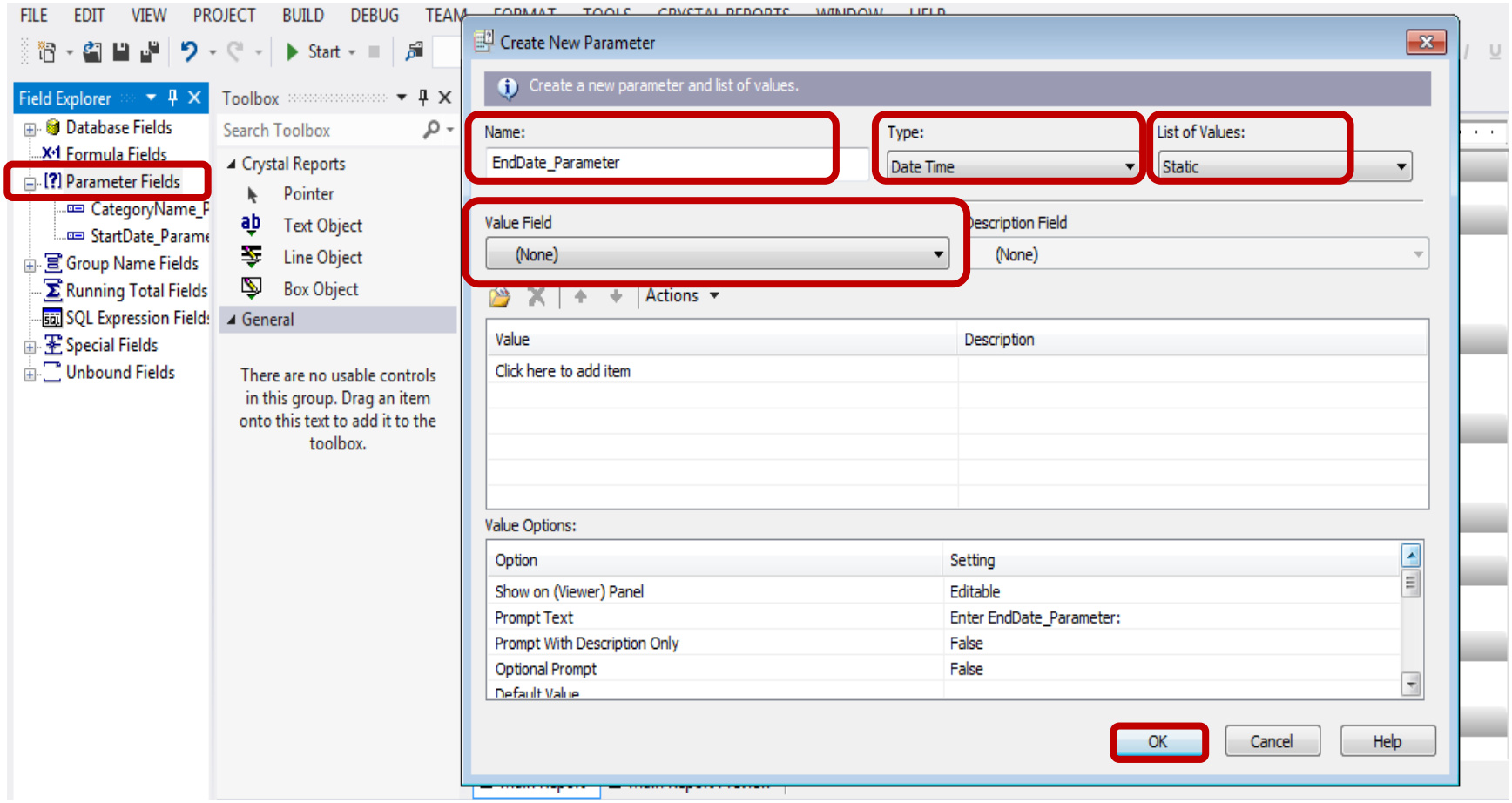
Crystal Reports with parameters(cont'd)



Crystal Reports with parameters(cont'd)



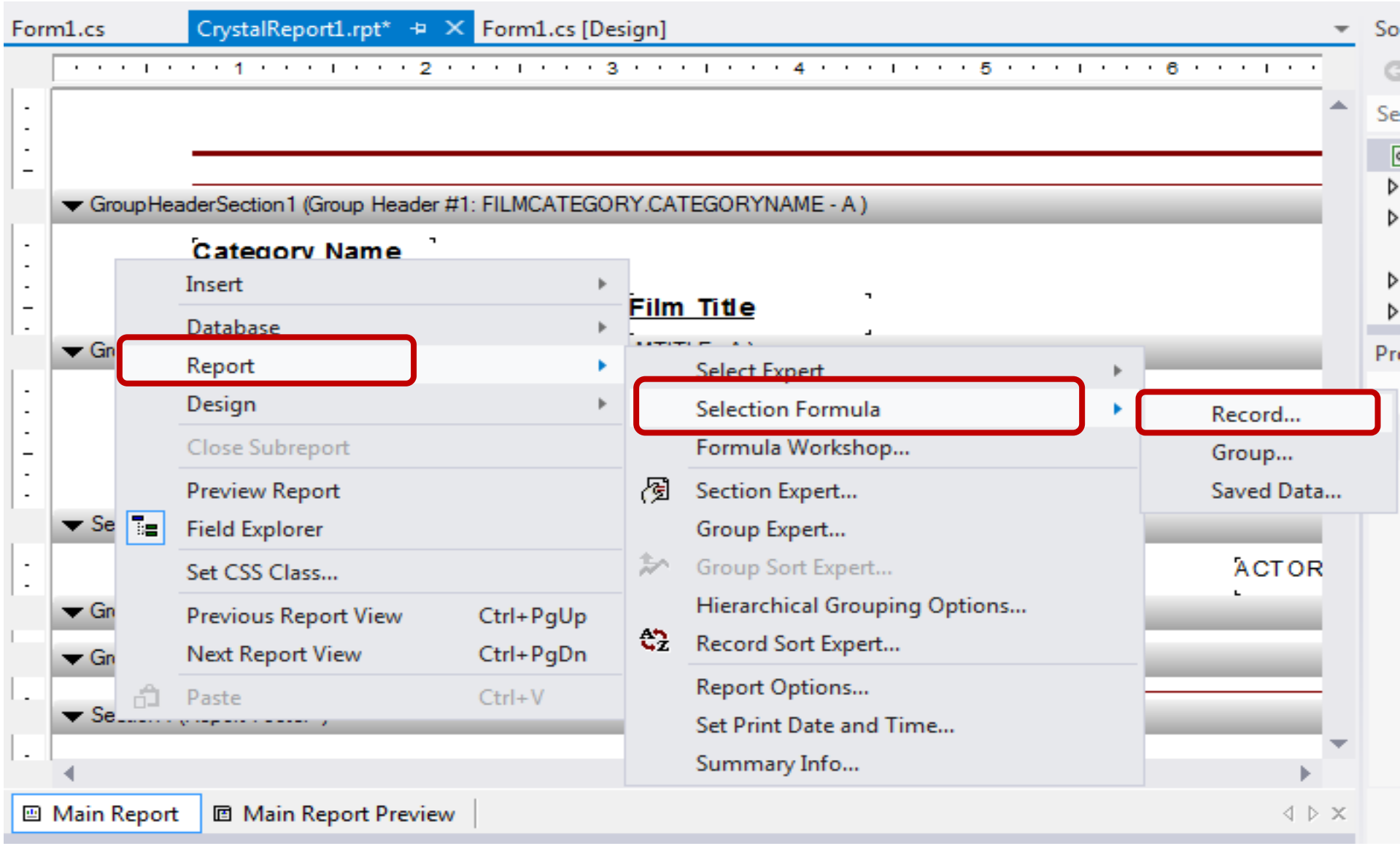
Crystal Reports with parameters(cont'd)

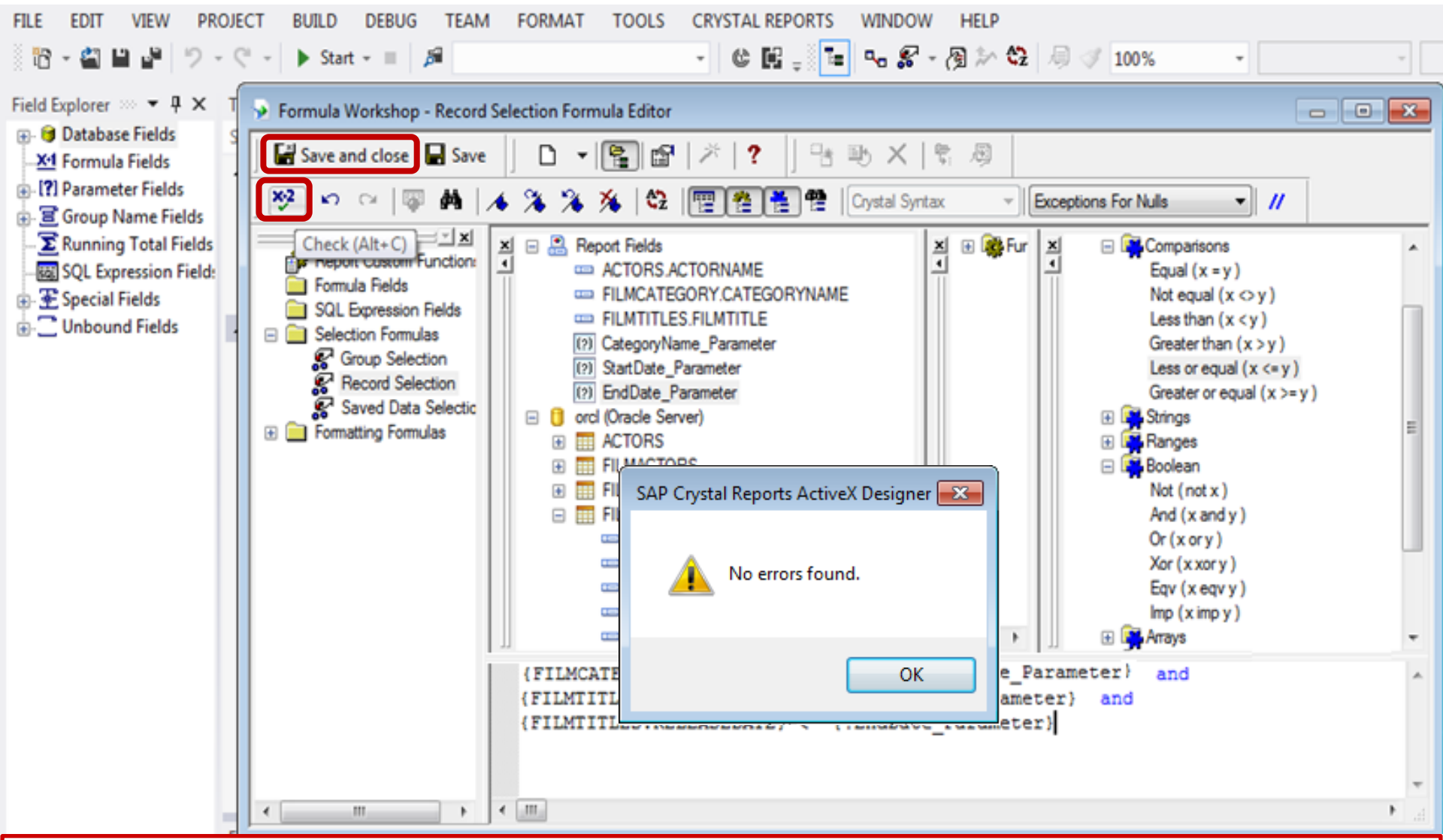


Crystal Reports with parameters(cont'd)

- **Name**
  - The parameter name which will be linked to a table column to filter the report data.
- **Type**
  - Parameter data type
  - It must match the data type of the column that will be linked to the parameter to filter the data
- **List of values**
  - Add a pre-defined list of values to the parameter so that user can choose among these values
  - Static means the values in the list will never change.
- **Value field**
  - Fill the list with a set of values found in a certain column in the database
- **Append all database values**
  - Add all values which exist currently in the selected column to the list of values of the parameter

After creating the parameter, it should be linked to a database column in a condition to filter the report data





{FilmCategory.CategoryName} = {?CategoryName_Parameter} and
{FilmTitles.ReleaseDate} >= {?StartDate_Parameter} and
{FilmTitles.ReleaseDate} <= {?EndDate_Parameter}

Enter Values

Enter Category_name_parameter — Category_name_parameter

...

Enter a Value:

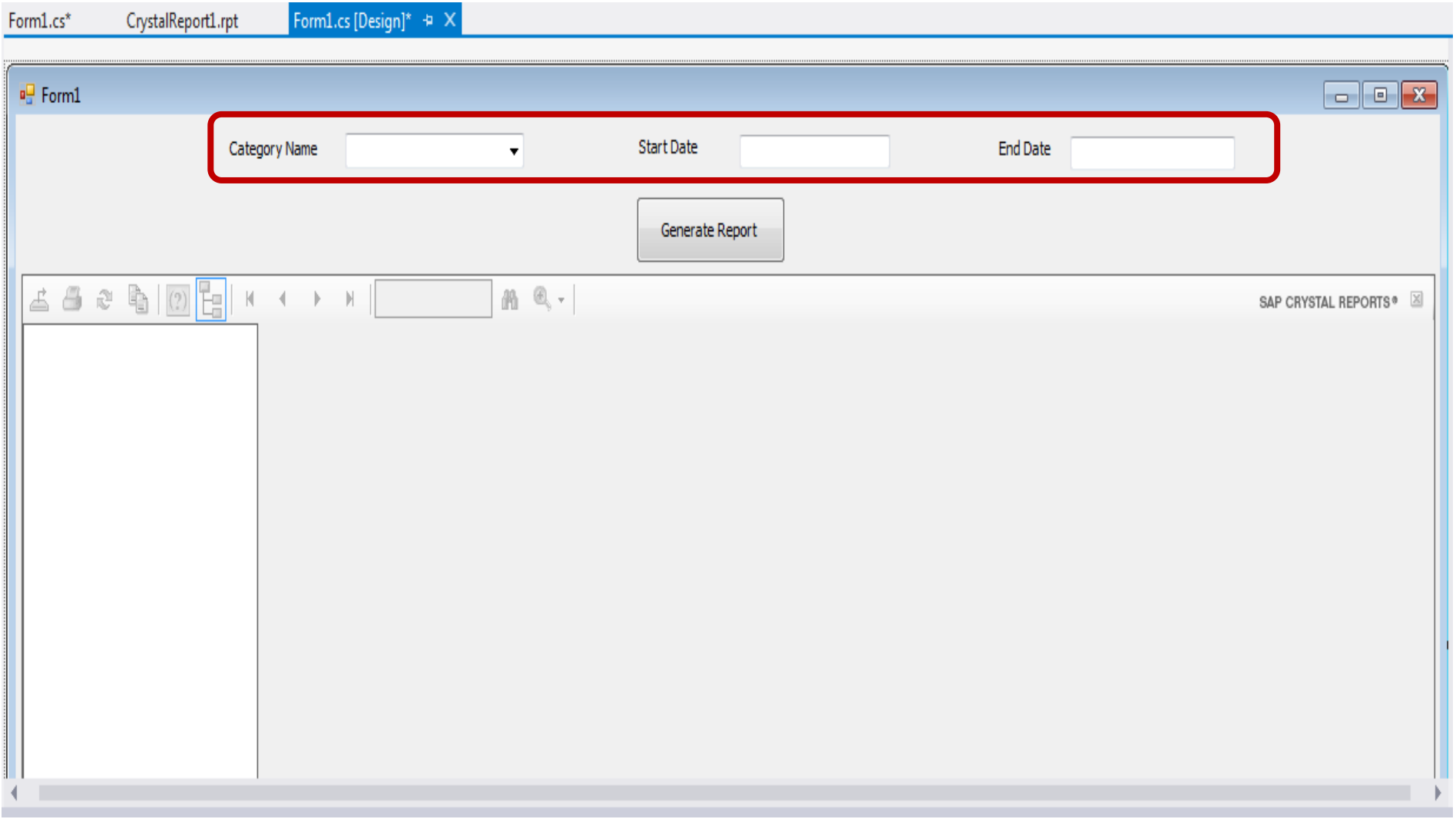
Enter Start_date_parameter: — Start_date_parameter

Enter End_date_parameter: — End_date_parameter

OK | Cancel

Main Report | Main Report Preview

Passing Parameter Values From C# Form to the Crystal Report


Form1.cs*
CrystalReport1.rpt
Form1.cs [Design]*
Form1
Category Name
Start Date
End Date
Generate Report
SAP CRYSTAL REPORTS®

Form1.cs* CrystalReport1.rpt*

CrystalReports_Lab1.Form1 Form1_Load(object sender,

```
using System.Windows.Forms;
using CrystalDecisions.Shared;

namespace CrystalReports_Lab
{
    public partial class Form1 : Form
    {
        CrystalReport1 CR;
        public Form1()
        { InitializeComponent(); }

        private void Form1_Load(object sender, EventArgs e)
        {
            CR = new CrystalReport1();
        foreach (ParameterDiscreteValue v in CR.ParameterFields[0].DefaultValues )
            Category_cmb.Items.Add(v.Value);
        }

        private void GenerateReport_btn_Click(object sender, EventArgs e)
        {
            CR.SetParameterValue(0, Category_cmb.Text);
            CR.SetParameterValue(1, Convert.ToDateTime (StartDate_txt.Text ));
            CR.SetParameterValue(2, Convert.ToDateTime (EndDate_txt.Text ));
            crystalReportViewer1.ReportSource = CR;
        }
```

Form1

Category Name Sci-Fi

Start Date 01-JAN-13 End Date 30-DEC-14

Generate Report

1 /1

SAP CRYSTAL

Sci-Fi

Main Report

Category Name

Sci-Fi

Film Title

Gravity

Actor Name

George Clooney

Sandra Bullock

Interstellar

Actor Name

Anne Hathaway

Jessica Chastain

Matthew McConaughey

Thank You